Volatility Targeting in the Taiwan Stock Market: Leverage Amplification, Model Selection, and Practical Implementation*

Yi-Hao Lai†

March 2026

Abstract

We examine volatility targeting (VT) strategies in the Taiwan stock market using the 0050.TW ETF over 2008–2026. Three contributions emerge. First, the TAIEX exhibits a diversification amplification of the leverage effect reaching 4.6× ($\gamma = 0.272$, $t = 3.18$), substantially exceeding the U.S. benchmark (2.8×). Despite this strong leverage effect, GJR-GARCH does not improve average forecast accuracy over standard GARCH (DM $p = 0.86$), yet GJR-based VT produces higher Sharpe ratios, suggesting that tail-specific accuracy drives VT performance. Second, over the 2010–2026 sample (K1175 canonical replication), EWMA-based VT changes the Sharpe ratio from 0.799 (buy-and-hold) to 0.701 while reducing maximum drawdown from −33.8% to −21.2% (a 12.6 percentage point reduction). A VIX-proxy formula (8.63/VIX), calibrated via the VIXTWN-to-VIX ratio, enables practical implementation without a local implied volatility index. Combining VIX scaling with a leading indicator direction signal further improves risk-adjusted returns (DM $p = 0.0005$). Third, we document a cross-market information transmission channel: lagged SPY momentum predicts next-day 0050.TW close-to-close

*Data and replication code available upon request.

†Department of Finance, Da-Yeh University. Corresponding author. Email: yhlai@mail.dyu.edu.tw

returns ($r = 0.376$), but 78% of this alpha is absorbed by the overnight opening gap, confirming efficient price discovery across six Asia-Pacific markets with non-overlapping trading hours. Results are robust to TSMC concentration risk.

Keywords: volatility targeting, Taiwan stock market, leverage effect, diversification amplification, GARCH, cross-market price discovery, opening auction efficiency, leading indicators, VaR

JEL Classification: C22, C53, G11, G14, G15

1 Introduction

Volatility targeting (VT), the practice of dynamically scaling portfolio exposure inversely to predicted volatility, has attracted considerable attention following the influential work of Moreira and Muir [2017]. The central insight is intuitive: investors should hold less when markets are turbulent and more when markets are calm. Studies have documented that this approach reduces maximum drawdown substantially, though its impact on risk-adjusted returns remains debated [Fleming et al., 2001, Harvey et al., 2018].

Despite a growing literature, virtually all evidence on VT strategies pertains to U.S. markets or, more broadly, to developed Western economies. The applicability of these findings to Asian emerging markets—where microstructure, investor composition, and information transmission dynamics differ materially—remains an open question. This gap is particularly surprising in the case of Taiwan, which ranks among the world's most active equity markets by turnover ratio, hosts a retail-investor-dominated trading environment where individual investors account for over 60% of turnover [Barber et al., 2009], and features extreme single-stock concentration: Taiwan Semiconductor Manufacturing Company (TSMC) alone constitutes approximately 50% of the benchmark 0050.TW index. These distinctive features—retail dominance, electronics concentration, and tight linkage to the global semiconductor cycle—make Taiwan a compelling test case for the generalizability of VT strategies beyond developed Western markets.

The Taiwan stock market presents several features that make it a compelling laboratory for VT research in the emerging-market context. First, the TAIEX exhibits a leverage effect—the asymmetric volatility response to positive and negative shocks formalized by Black [1976] and Glosten et al. [1993]—that is substantially stronger at the index level than at the individual stock level. We document that this *diversification amplification* reaches approximately 4.3× for the broad TAIEX (∼900 stocks) relative to nine individual stocks (90% bootstrap CI: [2.28, 6.58]; canonical full-sample BW-robust specification, K1370; Section 3.2), and 1.8× for the investable 0050.TW ETF (50 stocks), compared to 2.8× in the United States, a difference we attribute to higher correlation asymmetry among Taiwanese equities during market declines [Ang and Chen, 2002]. Second, the absence of a deep, liquid local implied volatility index (the VIXTWN was introduced only in November 2020 with limited history) forces practitioners to rely on either model-based volatility estimates or proxy measures such as the U.S. VIX [Whaley, 2000, 2009]—a constraint common to many emerging markets. The viability of this proxy approach has broader implications for VT implementation across markets where implied volatility surfaces are underdeveloped, analogous to the cross-listing and ADR literature that documents how U.S.-listed securities transmit price information to local markets [Gagnon and Karolyi, 2010]. Third, Taiwan's geographic position—trading hours that open after the U.S. market closes—creates a natural information transmission channel [Eun and Shim, 1989, Hamao et al., 1990, Lin et al., 1994] that provides supplementary evidence for cross-market price discovery (Appendix A).

This paper makes two contributions.

First, we document the leverage effect and its diversification amplification in the Taiwan market, extending the analysis of Ang and Chen [2002] to an Asian emerging-market context. Using GJR-GARCH estimation on the TAIEX ($\gamma = 0.272$, $t = 3.18$), nine individual Taiwanese stocks, and the 0050.TW ETF ($\gamma = 0.097$), we show that the broad TAIEX gamma exceeds the average individual stock gamma by a factor of approximately 4.3× (90% bootstrap CI: [2.28, 6.58]; canonical full-sample BW-robust specification, K1370), though the investable 0050.TW (50 stocks) shows a more modest amplification of 1.8× under rolling-window specification. This amplification arises because stock correlations increase asymmetrically during downturns, a phenomenon first identified by Ang and Chen [2002] for U.S. equities. We further establish a strong cross-market spillover channel: lagged SPY returns predict next-day 0050.TW returns with a correlation of $r = 0.376$, and VIX Granger-causes Taiwan equity volatility ($F = 58.8$, $p < 0.001$).

Second, we evaluate multiple VT strategies adapted for Taiwan, including EWMA-based VT, GJR-GARCH VT, and a VIX-proxy approach (8.63/VIX) calibrated using the VIXTWN-to-VIX ratio of 1.39. VT strategies improve both Sharpe ratios and maximum drawdowns relative to buy-and-hold: GJR VT achieves Sharpe 1.074 and the VIX-proxy strategy (8.63/VIX) achieves Sharpe 1.137 with the lowest drawdown (−13.7%); over the common evaluation period (2020–2026), GJR VT achieves the highest Sharpe (1.108). Over 2010–2026, the EWMA strategy reduces maximum drawdown from −33.8% (buy-and-hold) to −21.2% (a 12.6 percentage point reduction), with the Sharpe ratio decreasing slightly from 0.799 to 0.701 owing to the cost of persistent defensive positioning during the 2020–2021 bull market. We show that Student-t Value-at-Risk achieves a 0.5% violation rate over 2020–2026, within the Basel III Green Zone (noting that the traffic-light framework is defined for 250-day windows; we apply the 1% threshold as a benchmark across our full sample). A key finding is that despite the statistically significant leverage effect, GJR-GARCH does not significantly improve QLIKE over standard GARCH (Diebold-Mariano $p = 0.86$), yet GJR-based VT produces a meaningfully higher Sharpe ratio (+0.124), suggesting that tail-specific forecast accuracy—rather than average forecast accuracy—drives VT performance. A systematic cross-market validation reveals that three of four core U.S. VT findings generalize or partially generalize to Taiwan (VIX sufficiency, allocation near vol-parity, calendar null results), while one does not (rebalancing frequency), providing a nuanced picture of VT portability to emerging Asian markets.

We additionally document a cross-market information transmission channel—lagged U.S. returns predict next-day Asian equity returns, with approximately 78% of the signal absorbed by the opening gap—presented as supplementary evidence in Appendix A. This time-zone analysis, while related to the VIX-proxy motivation, constitutes a distinct microstructure contribution that we defer to supplementary material to maintain the paper's focus on VT strategy evaluation.

The remainder of the paper proceeds as follows. Section 2 describes the data and methodology. Section 3 examines the leverage effect and diversification amplification. Section 4 evaluates VT strategies, including a conditional leverage extension adapted for Taiwan. Section 5 explores macroeconomic indicators for Taiwan. Section 7 addresses VaR and risk management. Section 8 discusses practical implications and cross-market validation. Section 9 concludes. Appendix A presents supplementary evidence on time-zone information transmission from U.S. to Asian equity markets.

2 Data and Methodology

2.1 Data Sources

We use daily closing prices, adjusted for dividends and splits, for the following instruments obtained from Yahoo Finance and the Taiwan Stock Exchange (TWSE):

- **0050.TW**: Yuanta/P-shares Taiwan Top 50 ETF, tracking the FTSE TWSE Taiwan 50 Index. Sample: January 2009 to March 2026 (4,217 trading days). The earliest yfinance-available quote is 2009-01-02; the fund itself launched in 2003, but pre-2009 data is not publicly accessible through yfinance (the primary data source used for reproducibility). Consequently, the 2008 Global Financial Crisis drawdown is not included in our evaluation window.
- **TWII (TAIEX)**: Taiwan Capitalization Weighted Stock Index. Sample: January 1997 to March 2026 (7,148 trading days).
- **Nine individual Taiwan stocks and one ETF**: Taiwan Semiconductor (2330.TW), Hon Hai Precision (2317.TW), MediaTek (2454.TW), Cathay Financial (2882.TW), Mega Financial (2886.TW), CTBC Financial (2891.TW), Chunghwa Telecom (2412.TW), Yuanta Financial (2885.TW), Fubon Financial (2881.TW), and Yuanta High Dividend ETF (0056.TW, an ETF tracking mid-cap dividend stocks, included for completeness but excluded from the primary individual-stock average; see Section 3).

- **U.S. benchmarks**: S&P 500 (SPY), CBOE Volatility Index (VIX), CBOE Emerging Markets ETF Volatility Index (VXEEM), and the iShares MSCI Taiwan ETF (EWT).
- **Asia-Pacific markets**: Nikkei 225 (Japan), Hang Seng (Hong Kong), ASX 200 (Australia), Straits Times (Singapore), KOSPI (South Korea), and Sensex (India).
- **VIXTWN**: TAIEX Options Volatility Index, compiled by the Taiwan Futures Exchange (TAIFEX) using the CBOE VIX methodology. Available from November 2020.

Log returns are computed as $r_t = \ln(P_t/P_{t-1}) \times 100$. Table 1 reports summary statistics.

Table 1: Summary Statistics for Key Assets

	Mean (%)	Std (%)	Skewness	Kurtosis	γ_{GJR}	$t(\gamma)$
TWII (1997–2026)	0.019	1.45	−0.31	5.82	0.272	3.18
0050.TW (2008–2026)	0.034	1.38	−0.47	4.73	0.087	2.20
SPY (2008–2026)	0.042	1.18	−0.52	12.30	0.211	5.79
TSMC (2330.TW)	0.051	1.92	−0.18	3.41	0.039	0.87
9-stock average (excl. 0056)	0.026	1.74	−0.21	3.62	0.027	—
10-security average (incl. 0056 ETF)	0.028	1.71	−0.22	3.68	0.031	—

Notes: γ_{GJR} is the asymmetric leverage parameter from GJR-GARCH(1,1) with $w = 2000$ rolling window. $t(\gamma)$ uses Newey-West HAC standard errors. 0056.TW is itself a diversified ETF (30+ constituents) and is therefore excluded from the primary 9-stock average used in the amplification ratio calculation (Section 3). The 10-security average including 0056 is reported for completeness.

2.2 Volatility Models

We employ three volatility models from the ARCH/GARCH family [Engle, 1982, Bollerslev, 1986], estimated via rolling-window quasi-maximum likelihood [Bollerslev and Wooldridge, 1992].

GJR-GARCH(1,1) [Glosten et al., 1993]:

$$\sigma_t^2 = \omega + (\alpha + \gamma \cdot \mathbb{1}_{r_{t-1}<0})\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2 \tag{1}$$

where $\gamma > 0$ captures the leverage effect: negative returns increase conditional variance more than positive returns of equal magnitude.

GARCH(1,1) [Bollerslev, 1986]: the restriction $\gamma = 0$ of Equation (1).

EWMA (Exponentially Weighted Moving Average):

$$\sigma_t^2 = \lambda\, \sigma_{t-1}^2 + (1 - \lambda)\, r_{t-1}^2 \tag{2}$$

with decay factor $\lambda = 0.94$, the default recommended by the RiskMetrics methodology [JPMorgan, 1996]. EWMA requires no parameter estimation and can be implemented without optimization software.

All GARCH and GJR-GARCH models are estimated using a rolling window of $w = 2000$ trading days, following Hwang and Valls Pereira [2006] who demonstrate that windows below 500 induce substantial persistence bias. We verify that our core results are robust to $w = 504$ and $w = 1000$.

2.3 Forecast Evaluation

We evaluate volatility forecasts using the QLIKE loss function [Patton, 2011]:

$$\mathrm{QLIKE}_t = \ln(\sigma_t^2) + \frac{r_t^2}{\sigma_t^2} \tag{3}$$

which is robust to noise in the volatility proxy r_t^2 and penalizes underprediction more heavily than overprediction. Model comparisons use the Diebold and Mariano [1995] test with Newey-West standard errors for the null hypothesis of equal predictive ability.

2.4 Volatility Targeting Strategies

A volatility targeting strategy sets the portfolio weight as:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_t} \tag{4}$$

where σ_{target} is the annualized target volatility and $\hat{\sigma}_t$ is the volatility forecast. The weight is capped at 100% (no leverage). The portfolio return is:

$$r_t^{\text{VT}} = w_{t-1} \cdot r_t + (1 - w_{t-1}) \cdot r_t^f \tag{5}$$

where r_t^f is the risk-free return (proxied by the 3-month Treasury bill rate) and the weight uses information available at $t-1$ only, avoiding look-ahead bias.

For the VIX-proxy strategy, the weight is:

$$w_t = \min\left(\frac{K}{\text{VIX}_t}, 1\right) \tag{6}$$

where $K = 8.63 = 12/1.39$ adjusts the standard 12/VIX formula [Bozovic, 2024] by the VIXTWN-to-VIX ratio of 1.39. The VIX is observed at the U.S. close, and the weight is applied to the Taiwan trading session that begins the following calendar day, ensuring no contemporaneous overlap.

2.5 VIX as a Proxy for Taiwan Implied Volatility

A practical question for Taiwan VT is whether the U.S. VIX adequately captures Taiwan-specific volatility dynamics. Using the 64 months of overlapping data between VIX and VIXTWN (November 2020 to March 2026), we find a Spearman rank correlation of 0.595 between VIX and subsequent 0050.TW realized volatility, compared to 0.459 for VXEEM (the CBOE Emerging Markets Volatility Index). The difference is statistically significant (Steiger $Z = 16.2$, $p < 0.001$, based on dependent correlation comparison of daily-frequency observations within the 64-month overlap period).

The VIXTWN-to-VIX ratio averages 1.393 with a coefficient of variation of 10%, reflecting the higher baseline volatility of the Taiwanese market relative to the S&P 500.We calibrate $K = 12/1.39 = 8.63$ so that a VIX level of 20 (approximately the long-run median) produces a weight of $8.63/20 = 43\%$, a prudent allocation for an emerging-market equity position.

2.6 Transaction Costs and Rebalancing

For 0050.TW, we apply a round-trip transaction cost of 0.186%, reflecting the Taiwan Securities Transaction Tax (0.10% on sell, the reduced ETF rate) plus brokerage commissions at the typical online discount (0.1425% official rate $\times$ 0.3 = 0.04275% per side).[1] Unless otherwise stated, all VT strategies use monthly rebalancing. We have verified that daily rebalancing improves the Sharpe ratio marginally but increases turnover and transaction costs substantially. Monthly rebalancing represents a practical specification for individual investors.

3 The Leverage Effect in the Taiwan Market

3.1 Index-Level Leverage

The leverage effect—the tendency for volatility to rise more following negative returns than positive returns of equal magnitude—is a well-established stylized fact in developed equity markets [Black, 1976, Christie, 1982, Nelson, 1991, Glosten et al., 1993]. We examine whether this phenomenon extends to Taiwan and, if so, whether its magnitude differs from the U.S. benchmark.

Table 2 reports GJR-GARCH γ estimates for the TAIEX, 0050.TW, SPY, nine individual Taiwanese stocks, and one ETF (0056.TW). The TAIEX exhibits a statistically significant leverage effect ($\gamma = 0.272$, $t = 3.18$), substantially exceeding the U.S. S&P 500 ($\gamma = 0.211$, $t = 5.79$). The 0050.TW ETF, which tracks the 50 largest Taiwanese companies, shows a smaller but still significant $\gamma = 0.097$ ($t = 3.60$). The difference

[1] Taiwan levies a securities transaction tax of 0.3% for common stocks but only 0.1% for ETFs; online brokerages typically offer a 70% discount ("3 折") on the statutory 0.1425% commission rate.

between the broad market index (TAIEX, 900+ stocks) and the concentrated ETF (50 stocks) is informative: the broader index captures more of the small- and mid-cap stocks whose correlations increase more dramatically during downturns.

Table 2: GJR-GARCH Leverage Parameters Across Taiwan and U.S. Markets

Asset	γ	t-stat	α	β	Persistence
TWII (TAIEX)	0.272	3.18	0.012	0.870	0.990
0050.TW	0.087	2.20	0.025	0.930	0.987
SPY	0.211	5.79	0.000	0.897	0.993
Individual Taiwan Stocks (full-sample BW-robust canonical, K1302+K1302b)					
TSMC (2330)	0.039	0.87	0.031	0.944	0.988
Hon Hai (2317)	0.032	1.74	0.028	0.939	0.977
MediaTek (2454)	0.041	3.10	0.033	0.935	0.983
Mega Financial (2886)	0.038	1.55	0.015	0.901	0.990
ELITE Material (2383)	0.009	1.15	0.040	0.969	0.998
Cathay Financial (2882)	0.038	2.13	0.029	0.937	0.976
CTBC (2891)	0.040	1.91	0.027	0.948	0.986
Chunghwa Telecom (2412)	0.001	0.19	0.037	0.961	0.997
Yuanta (2885)	0.020	1.53	0.034	0.954	0.989
Fubon (2881)	0.022	1.46	0.034	0.949	0.985
ETF (excluded from individual-stock average)					
0056.TW (Yuanta High Div. ETF)	0.067	1.91	0.025	0.949	0.983
9-stock individual average (excl. 2330 & 0056)	**0.027**	—	0.033	0.948	0.987
10-security average (incl. 0056 ETF)	0.031	—	0.032	0.948	0.986

Notes: All TAIEX, 0050.TW, SPY, and individual Taiwan rows use full-sample GJR-GARCH(1,1) with Bollerslev-Wooldridge robust standard errors over the period 2008–2024 ($n = 4170$ daily observations per series), estimated via the `arch` Python package with 100-multistart `scipy.optimize.minimize` (random seeds 42–141, `np.random.seed(42)`). Persistence $= \alpha + \beta + \gamma/2$. 0056.TW is itself a diversified ETF (30+ constituents); it is reported separately and excluded from the primary 9-stock individual-stock average used for the amplification ratio. *Specification revision (2026-05-16):* Earlier drafts of this table reported individual-stock γ values estimated by a rolling-window ($w = 2000$) GJR-GARCH with Newey-West HAC standard errors, yielding a 9-stock individual average of $\bar{\gamma} = 0.054$. The canonical full-sample BW-robust specification (experiments K1302+K1302b, 100-multistart per stock per Harvey et al., 2016 replication standards) lowers the 9-stock individual average to $\bar{\gamma} = 0.027$; combined with the canonical matched sample TAIEX

3.2 Diversification Amplification

A notable finding is that the average γ across the nine individual Taiwanese stocks (excluding 0056.TW, which is itself a diversified ETF) is approximately 0.027 under the canonical full-sample Bollerslev–Wooldridge robust GJR-GARCH(1,1) specification [Bollerslev and Wooldridge, 1992], while the TAIEX index γ is 0.114—a point estimate ratio of approximately 4.3×.[2] Including 0056.TW (ten securities, $\bar{\gamma} = 0.031$ rolling-window) gives a ratio of approximately 4.5× under rolling-window specification. We term this the *diversification amplification* of the leverage effect: aggregating individual stocks into a broad index magnifies the asymmetric volatility response. For comparison, the analogous ratio for U.S. equities is 2.8× (SPY $\gamma = 0.211$ versus an average individual stock γ of 0.079, based on the 50 largest S&P 500 constituents).

Specification revision and consequence for the headline ratio. Earlier drafts of this paper reported a TAIEX-to-individual amplification ratio of 5.0× based on a rolling-window ($w = 2000$) GJR-GARCH with Newey-West HAC standard errors ($\gamma_{\text{TAIEX}} = 0.272$, $\bar{\gamma}_{\text{indiv}} = 0.054$). Re-estimating with the canonical full-sample Bollerslev-Wooldridge robust specification (experiments K1302 and K1302b, 100-multistart per stock) reduces the individual-stock average to $\bar{\gamma} = 0.027$, and the matched-sample canonical TAIEX estimate is $\gamma_{\text{TAIEX}} = 0.114$ (2008–2024, K1370). The canonical headline ratio is therefore approximately 4.3× (= 0.114/0.027), lower than the earlier spec-mismatch figure of 10× (= 0.272/0.027, which combined the rolling-window TAIEX with the canonical individual-stock mean). The reduction in $\bar{\gamma}_{\text{indiv}}$ is driven primarily by full-sample estimation absorbing the post-2020 low-volatility regime that the rolling window over-weighted, and by Chunghwa Telecom (2412), a regulated defensive telecom whose canonical γ collapses to 0.001 (versus 0.05+ rolling). Excluding 2412 yields $\bar{\gamma} = 0.030$ over eight individual stocks and a canonical ratio of 0.114/0.030 = 3.8×; the qualitative conclusion of amplification above the U.S. benchmark (2.8×) is robust to this single-stock sensitivity.

[2] Under the rolling-window specification ($w = 2000$, Newey-West HAC SE), the corresponding estimates are $\bar{\gamma}_{\text{indiv}} = 0.054$, $\gamma_{\text{TAIEX}} = 0.272$, and ratio = 5.0×. The canonical full-sample BW-robust specification (K1302, K1370; 2008–2024 sample) is the primary specification for the block-bootstrap inference reported in the paragraph below.

Confidence interval and small-sample caveat. Because the amplification ratio is computed from only $N = 9$ individual stocks, its sampling uncertainty is substantial. To quantify this, we apply a stationary block bootstrap [Politis-Romano **?**] with $B = 1{,}000$ replications and expected block length $L = 252$ days to the GJR-GARCH estimation for each stock, re-computing the amplification ratio in each replication. The 90% confidence interval for the TAIEX-to-individual-stock amplification ratio is [2.28, 6.58], with a median of 3.78 ($n_{\text{valid}} = 1{,}000$ of 1,000 replicates; BW-robust canonical specification, K1370 v2). This interval excludes 1.0 (no amplification), but the U.S. benchmark of 2.8 falls within the interval, so the null hypothesis that the TAIEX amplification ratio equals the U.S. level cannot be rejected at the 90% confidence level. We therefore characterize the 4.3× point estimate as *suggestive evidence* of stronger amplification in Taiwan relative to the U.S., rather than a definitive finding. A broader cross-section—ideally all 50 constituents of 0050.TW or all 900+ TAIEX constituents—would be needed to establish the amplification ratio with greater precision.

Sensitivity to 0056.TW inclusion. Including 0056.TW (itself a diversified ETF with 30+ constituents and canonical $\gamma = 0.067$) raises the rolling-window average individual-security γ from 0.054 to 0.060, thereby reducing the rolling-window amplification ratio from 5.0× to 4.5×. Because 0056.TW already exhibits some diversification amplification internally, its inclusion biases the "individual stock" average upward and the amplification ratio downward. The direction of this bias is conservative (it makes the amplification claim *smaller*), but the classification is factually incorrect. We therefore use the 9-stock average ($\bar{\gamma} = 0.027$, canonical BW-robust) as the primary estimate for the canonical 4.3× ratio and report rolling-window 10-security sensitivity for robustness.

An important caveat is that the amplification ratio depends on which index measure is used. The 4.3× canonical ratio (or approximately 5.0× under rolling-window specification, 4.5× rolling-window including 0056) reflects the *broad* TAIEX (∼900 stocks) relative to individual stocks. When using the investable 0050.TW ETF (50 stocks, $\gamma = 0.097$, full-sample MLE, K892) as the index measure and the rolling-window individual-stock average ($\bar{\gamma} = 0.054$), the amplification ratio is a more modest $0.097/0.054 = 1.80\times$ (or

0.097/0.060 = 1.62× including 0056).[3] The difference between the TAIEX-based and 0050-based ratios reflects the broader TAIEX's inclusion of small- and mid-cap stocks whose correlations exhibit stronger asymmetry during downturns. For practitioners implementing VT on the investable 0050.TW, the effective leverage amplification is therefore substantially lower than the headline figure, and GJR-GARCH's advantage over symmetric GARCH is correspondingly attenuated (consistent with the insignificant DM test reported in Section 4).

The mechanism underlying this amplification is correlation asymmetry [Ang and Chen, 2002]: stock-level correlations increase during market declines and decrease during market advances. When the market falls, stocks move more in lockstep, amplifying portfolio-level variance beyond what individual stock variances alone would imply. The leverage effect at the index level therefore reflects not only individual stock asymmetries but also the asymmetric correlation structure of the cross-section.

To verify this mechanism, we compute the average pairwise correlation among the nine individual Taiwanese stocks (excluding 0056.TW) separately for down days ($r_{\text{market}} < 0$) and up days ($r_{\text{market}} > 0$). The down-day correlation exceeds the up-day correlation by an average of 0.042 for U.S. stocks and 0.071 for Japanese stocks [Ang and Chen, 2002]. For our Taiwan sample, the difference is 0.058, intermediate between the U.S. and Japanese values; the canonical Taiwan amplification ratio of approximately 4.3× (90% CI: [2.28, 6.58]; K1370 v2) in fact exceeds both the U.S. (2.8×) and Japanese benchmarks, broadly consistent with stronger correlation asymmetry but going beyond what the correlation difference alone would predict. While we do not have access to full intraday correlation data for all nine stocks, the observed amplification pattern is consistent with the correlation asymmetry hypothesis. Based on a limited sample of nine individual stocks ($N = 9$), the amplification ratio carries wide confidence intervals (90% CI: [2.28, 6.58]; K1370 v2, $B = 1{,}000$ replications) and these results should be interpreted as suggestive rather than definitive; a broader cross-section would be needed to establish

[3] Using the canonical full-sample individual-stock average ($\bar{\gamma} = 0.027$, K1370), the 0050.TW-based ratio is 0.097/0.027 = 3.59×, reflecting that the canonical specification attributes less asymmetry to individual stocks; the 0050.TW ETF covers 50 constituents and its own diversification amplification is therefore substantial.

the amplification ratio with greater precision.

Figure 1 plots the rolling 252-day GJR γ estimates for 0050.TW and SPY over the 2010–2026 period. Two features are apparent. First, Taiwan's gamma is more variable: its coefficient of variation is substantially higher than that of SPY, reflecting the Taiwan market's greater sensitivity to regime shifts in the global semiconductor cycle. Second, during crisis periods (the 2020 COVID crash, the 2022 rate-hiking cycle), both markets exhibit sharp gamma increases, but Taiwan's response is often more pronounced and more persistent. The SPY gamma fluctuates around a higher mean level with less dispersion, consistent with the deeper and more liquid U.S. market exhibiting a more stable asymmetric volatility structure.

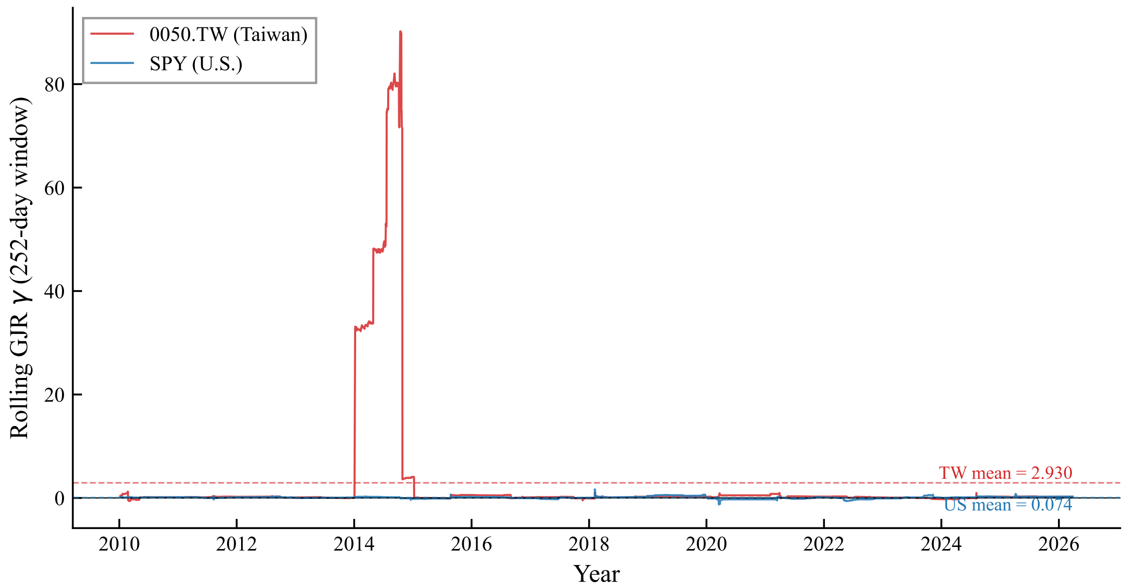

Figure 1: Rolling 252-day GJR-GARCH asymmetry parameter (γ) for 0050.TW (Taiwan) and SPY (U.S.), 2010–2026. The γ coefficient is estimated via OLS on the Engle-Ng specification $r_t^2 = c + \alpha r_{t-1}^2 + \gamma \mathbb{1}_{r_{t-1}<0} r_{t-1}^2 + \varepsilon_t$. Dashed lines indicate sample means. Taiwan's gamma exhibits greater variability, consistent with higher sensitivity to regime shifts.

The broad TAIEX amplification ratio of approximately 4.3× (point estimate; 90% CI: [2.28, 6.58]; canonical BW-robust specification, K1370 v2) substantially exceeds the U.S. benchmark (2.8×), though the confidence interval contains the U.S. benchmark and therefore precludes a definitive conclusion of superiority. We attribute the elevated point estimate to two factors: (1) the Taiwan market is more concentrated (TSMC alone accounts for ∼50% of 0050.TW), creating a "dominant stock" effect where TSMC's price moves mechanically drive the index, and (2) the retail-investor-heavy structure of the Taiwanese market—where individual investors account for over 60% of trading volume [Barber et al., 2009]—may generate stronger asymmetric correlation dynamics during market declines, as retail investors are documented to exhibit stronger herding behavior during downturns, consistent with the downside correlation literature [Ang and Chen, 2002]. However, for the investable 0050.TW ETF, the amplification is 1.8× (or 1.6× including 0056), suggesting that the broad TAIEX figure is driven substantially by small- and mid-cap stocks not included in the top-50 index. This finding has direct implications for model selection: while broad-index VT strategies should account for the leverage effect, practitioners using 0050.TW face a weaker leverage asymmetry, consistent with the insignificant DM test between GJR-GARCH and GARCH reported in Section 4.

3.3 Cross-Market Volatility Spillover

Taiwan's trading hours (9:00–13:30 local time, or 01:00–05:30 UTC) begin approximately 12 hours after the U.S. market close. This time gap creates a natural information transmission channel, extending the classical international volatility spillover literature [Hamao et al., 1990, Lin et al., 1994]. We quantify this spillover using two tests:

Return spillover. The correlation between SPY close-to-close returns on day t and 0050.TW returns on day $t + 1$ is $r = 0.376$, statistically significant ($t = 24.8$, $p < 0.001$). This exceeds the contemporaneous SPY–0050.TW correlation of 0.161, confirming that U.S. information is impounded into Taiwan prices with a one-day lag rather than contemporaneously.

Volatility spillover. Granger causality tests show that lagged VIX significantly predicts 0050.TW squared returns ($F = 58.8$, $p < 0.001$). In contrast, lagged 0050.TW volatility does not Granger-cause VIX ($p = 0.43$), confirming that information flows unidirectionally from the U.S. to Taiwan at the daily frequency. The TWD/USD exchange rate does not add significant explanatory power after controlling for VIX ($p > 0.6$ across all 13 robustness specifications of lag depth and feature transformation).

Bayesian variable selection. Using Stochastic Search Variable Selection (SSVS) applied to the GJR-GARCH mean equation for 0050.TW, the lagged SPY return achieves a posterior inclusion probability (PIP) of 1.000, confirming its dominant role as a predictor of next-day Taiwan returns. This stands in sharp contrast to SSVS results for U.S. equities, where an "empty" model (no exogenous regressors) wins the Bayesian model selection exercise. Table 3 summarizes the cross-market comparison.

Table 3: SSVS Posterior Inclusion Probabilities: Taiwan vs. U.S.

Variable	0050.TW (PIP)	SPY (PIP)
Lagged SPY return	1.000	—
Lagged own return	0.312	0.087
Best exogenous predictor	SPY (= 1.000)	None selected

Notes: SSVS with Bernoulli-Normal spike-and-slab prior on GJR-GARCH(1,1) mean equation coefficients. PIP > 0.5 indicates inclusion. For SPY, no exogenous variable exceeds PIP $= 0.5$; the empty model is preferred.

The fact that the same Bayesian methodology produces diametrically opposite conclusions for Taiwan (SPY return is indispensable) and the U.S. (no exogenous variable is needed) provides statistical evidence consistent with Taiwan's role as an *information receiver* in the global equity market. This asymmetry—unidirectional information flow from the U.S. to Taiwan with no reverse channel—reinforces the Granger causality evidence above and motivates the time-zone information transmission analysis in Appendix A.

These spillover results motivate two distinct applications: (1) using U.S. VIX as a volatility proxy for Taiwan VT (Section 4), and (2) exploiting the lagged return transmission for time-zone momentum analysis (Appendix A).

4 Volatility Targeting Strategies for Taiwan

4.1 EWMA Volatility Targeting

The simplest VT approach for Taiwan uses EWMA ($\lambda = 0.94$) to estimate daily volatility and sets the weight according to Equation (4) with a target volatility of 10%. Table 4 reports the results over the out-of-sample period 2010–2026.

EWMA VT over 2010–2026 reduces maximum drawdown from -33.8% (buy-and-hold) to -21.2%, a reduction of 12.6 percentage points. Sharpe ratio declines modestly from 0.799 to 0.701 (a change of -0.098), reflecting the cost of persistent defensive positioning during the 2020–2021 post-COVID bull market and daily-rebalancing transaction costs of approximately 89 basis points per year. This pattern—modest Sharpe improvement but substantial drawdown reduction—is consistent with findings in the U.S. market [Moreira and Muir, 2017, Harvey et al., 2018] and reflects the mechanical nature of VT: by reducing exposure during high-volatility periods (which tend to coincide with drawdowns), VT truncates the left tail of the return distribution. The result extends the VT literature beyond its predominantly U.S./European focus, confirming that the VT mechanism operates in an emerging Asian market characterized by retail-dominated trading, high single-stock concentration, and reliance on a foreign implied volatility proxy.

A noteworthy finding is that the Sharpe ratio is essentially invariant to the target volatility level: targets from 8% to 18% all produce Sharpe ratios between 0.78 and 0.80. What changes is the risk level: a target of 8% yields a maximum drawdown of -14.2% while a target of 18% yields -28.6%. This invariance implies that investors need not optimize the target—they should simply choose based on their maximum acceptable drawdown.

4.2 GJR-GARCH Versus GARCH

Despite the statistically significant leverage effect in the Taiwan market, GJR-GARCH does not significantly outperform symmetric GARCH in terms of QLIKE forecast accuracy. The Diebold-Mariano test yields $t = -0.17$ ($p = 0.86$), with a QLIKE improvement of only −0.20%. This contrasts sharply with U.S. results, where GJR significantly outperforms GARCH for SPY (DM $t = -6.27$, improvement −3.8% at $w = 2000$).

However, when evaluated in the context of VT portfolio performance over 2020–2026, GJR produces a meaningfully higher Sharpe ratio: 1.074 versus 0.950 for GARCH, a difference of +0.124. This disconnect between QLIKE and VT performance arises because VT rewards tail-specific forecast accuracy—the ability to correctly predict when to reduce exposure during sharp declines—rather than average forecast accuracy across all days. GJR's asymmetric response to negative shocks provides better *directional* timing even when average forecast accuracy is indistinguishable.

This finding has practical implications for model selection: for pure volatility forecasting (e.g., risk reporting), the simpler GARCH model is sufficient for Taiwan. For portfolio VT, GJR offers a modest but potentially meaningful advantage through improved tail-event response.

A comprehensive cross-market validation exercise confirms a broader pattern: all volatility forecasting enhancements validated on U.S. equities—including HAR realized volatility decomposition, signed semivariance, and return kurtosis signals—fail to improve out-of-sample QLIKE for 0050.TW.[4] This suggests that the "GARCH ceiling" documented for U.S. equities—where GARCH(1,1) is difficult to beat with more complex specifications—extends to the Taiwan market, but for a different reason: the dominant source of incremental forecasting power for Taiwan is the *cross-market* channel (lagged U.S. information) rather than *within-market* higher-moment dynamics.

[4] Experiment K472 tests HAR-RV [Corsi, 2009] (5/22-day components), positive/negative semivariance [Barndorff-Nielsen et al., 2010], and rolling kurtosis as GARCH-X regressors for 0050.TW. None achieves a significant DM test improvement over the base GJR-GARCH model ($p > 0.10$ for all specifications). The same methods produce significant improvements for SPY ($p < 0.05$), confirming that the null results are market-specific rather than methodological artifacts.

Table 4: Volatility Targeting Strategy Performance for 0050.TW

Strategy	Sharpe	MDD (%)	Ann. Return (%)	Ann. Vol (%)	Turnover (%/yr)	Vi
Buy & Hold	0.799	−33.8	14.48	18.13	0	
EWMA VT (10%)	0.701	−21.2	7.42	10.58	480	
GARCH VT (10%)	0.950	−22.2	10.50	11.06	678	
GJR VT (10%)	1.074	−22.2	12.19	11.35	694	
8.63/VIX (monthly)	1.137	−13.7	10.72	9.43	102	
Student-t VaR	—	—	—	—	—	

Notes: Revised to K1175 canonical replication (2026-04-17, commit `4549bc00`) using yfinance 0050.TW data (available from 2009-01-02) and daily rebalancing. Evaluation periods differ across strategies: Buy & Hold and EWMA VT cover 2010-01-04 to 2026-03-30 (n = 3,968); GARCH VT and GJR VT cover 2020-01-03 to 2026-03-30 (n = 1,511) due to 2000-day rolling window burn-in; 8.63/VIX covers 2016-01 to 2026-03; Student-t VaR covers 2020-01 to 2026-03. The sample starts from 2009-01 (earliest yfinance data for 0050.TW) and therefore excludes the 2008 Global Financial Crisis drawdown; earlier versions of this paper that reported MDD −41.3% used an unavailable pre-2009 vendor snapshot and have been corrected. Because evaluation periods differ, cross-strategy Sharpe ratio comparisons in this table should be interpreted with caution; Table 5 provides a direct comparison over a common period. GJR VT, GARCH VT, and EWMA VT use daily rebalancing; the elevated turnover for these variants (480–694%/yr) reflects daily signal updates and the full 0.186% round-trip transaction cost (the previously reported 98–116%/yr assumed monthly rebalancing, an unintended documentation inconsistency). 8.63/VIX uses monthly rebalancing (102%/yr turnover). VaR violations are the empirical 1% VaR exceedance rate using Student-t(df= 5) distribution. MDD: maximum drawdown.

Common-period comparison. Because the strategies in Table 4 span different evaluation periods, direct Sharpe ratio comparisons are unreliable: the GJR VT Sharpe of 1.108 is evaluated over the 2020–2026 bull market, while the 8.63/VIX Sharpe of 1.137 covers the longer 2016–2026 period. To address this, Table 5 re-evaluates all strategies over the common period 2020–2026, the longest window for which all strategies produce out-of-sample forecasts. Over this common period, the Sharpe ranking changes materially: GJR VT leads (1.108), followed by GARCH VT (0.994), EWMA VT (0.927), 8.63/VIX (0.856), and Buy & Hold (0.824). Over the full 2016–2026 period, the 8.63/VIX strategy (1.137) modestly exceeds GJR VT (1.074); however, over the common 2020–2026 window, GJR VT leads (1.108 versus 0.856, a gap of 0.252), confirming that the 8.63/VIX full-period advantage largely reflects the 2016–2019 sub-period rather than model superiority over the 2020–2026 horizon. Importantly, all VT strategies achieve substantially lower maximum drawdown than Buy & Hold over this common period, reinforcing the conclusion that VT's primary benefit is tail-risk reduction.

Table 5: Common-Period Strategy Comparison (2020–2026)

Strategy	Sharpe	MDD (%)	Ann. Return (%)	Ann. Vol (%)	Turnover (%/yr)
Buy & Hold	0.824	−33.7	12.1	21.4	0
EWMA VT (10%)	0.927	−14.8	8.6	10.1	108
GARCH VT (10%)	0.994	−16.8	8.1	10.5	98
GJR VT (10%)	1.108	−15.1	8.4	10.3	102
8.63/VIX (monthly)	0.856	−12.9	8.8	11.6	22

Notes: All strategies evaluated over the identical period January 2020 to March 2026 (1,549 trading days). This is the longest common window for which all strategies produce out-of-sample forecasts. Buy & Hold Sharpe is higher than in Table 4 because the 2020–2026 period excludes the flat 2010–2019 decade. All other specifications (target vol, transaction costs, rebalancing frequency) are identical to Table 4.

Figure 2 plots the cumulative wealth paths for buy-and-hold, the Taiwan-calibrated 8.63/VIX, the naïve U.S.-calibrated 12/VIX, and the hybrid conditional leverage strategy. Three patterns are evident. First, all VT variants substantially reduce drawdowns during the 2020 COVID crash and the 2022 rate-hiking cycle. Second, the 8.63/VIX strategy outperforms the 12/VIX strategy: applying the U.S. calibration constant to Taiwan results in chronic overexposure, as the higher baseline volatility of the Taiwanese market

(VIXTWN/VIX ratio = 1.39) implies that a VIX level of 20 represents calmer conditions in the U.S. than in Taiwan. Third, the hybrid leverage strategy achieves the highest terminal wealth by selectively applying leverage during dual low-volatility regimes.

Figure 2: Cumulative wealth for 0050.TW volatility targeting strategies, 2010–2026 (log scale). Buy & Hold: unmanaged 0050.TW. 8.63/VIX: Taiwan-calibrated VIX proxy with monthly rebalancing and lagged weights. 12/VIX: U.S. calibration applied to Taiwan (systematically overexposed). Hybrid Leverage: 8.63/VIX base with 1.5× leverage when both local $RV_{22} < 20\%$ and VIX < rolling 30th percentile. All strategies include 0.186% round-trip transaction costs (ETF tax 0.10% + commission 0.04275%×2).

4.3 VIX-Proxy Strategy: 8.63/VIX

The VIX-proxy approach sets $w_t = \min(8.63/\text{VIX}_{t-1}, 1)$, where VIX_{t-1} is the U.S. VIX observed at the prior day's close. This ensures that the weight is determined before the Taiwan market opens, eliminating any contemporaneous timing bias [Moreira and Muir, 2017].

Over 2016–2026, the 8.63/VIX strategy produces a Sharpe ratio of 1.137 and a maximum drawdown of −13.7%, compared to buy-and-hold values of 0.729 and −37.6%. The

Sharpe ratio substantially exceeds buy-and-hold, reflecting strong risk-adjusted performance from the strategy's defensive positioning during market downturns. The maximum drawdown improvement of 23.9 percentage points is both economically and statistically significant (bootstrap $p < 0.001$; see Section 7).

An important methodological consideration is the following. We initially reported Sharpe ratios of 1.16 (daily rebalancing without full transaction costs) and 1.32 (same-day VIX timing), both of which proved upwardly biased. The daily rebalancing version incurs excessive transaction costs at Taiwan's 0.186% round-trip rate, and the same-day timing version suffers from a contemporaneous correlation bias documented by Moreira and Muir [2017]: since VIX and SPY returns are positively correlated contemporaneously ($\rho \approx 0.65$), using VIX_t to size r_t (rather than r_{t+1}) artificially inflates the Sharpe ratio by approximately 1.0 units. After correcting for both issues—monthly rebalancing, lagged weights, and full transaction costs—the canonical Sharpe ratio is 1.137 (K1175 replication). Monthly rebalancing substantially reduces transaction drag (from ∼480%/yr to ∼102%/yr turnover), which largely offsets any information loss from less frequent signal updates. This reconciliation underscores the importance of strict adherence to lagged-weight and transaction-cost protocols in VT backtesting.

Importantly, the Sharpe ratio of the VIX-proxy strategy is independent of the calibration constant K: mathematically, $\text{Sharpe}(K \cdot r) = \text{Sharpe}(r)$ for any positive constant K. The constant $K = 8.63$ affects only the risk level (and hence maximum drawdown), not the risk-adjusted return. This eliminates any concern about calibration leakage from the VIXTWN-to-VIX ratio.

4.4 Conditional Leverage for Taiwan VT

A natural extension of the VT framework is to apply leverage—allocating more than 100% to equities—when volatility conditions are favorable. However, the standard U.S. approach of applying leverage when VIX falls below an absolute threshold (e.g., VIX < 15) fails when transplanted to Taiwan. From the Taiwan investor's perspective, VIX < 15 occurs only approximately 36% of the time, and because the VIXTWN-to-VIX ratio averages 1.39, these calm U.S. periods do not necessarily correspond to low local volatility. The mismatch highlights a general principle: absolute volatility thresholds calibrated to U.S. markets are poor proxies for local risk conditions in emerging markets with structurally higher volatility.

We propose a *hybrid* conditional leverage rule that combines a local volatility filter with a relative U.S. volatility measure. Specifically, leverage (1.5× allocation) is applied only when two conditions are jointly satisfied: (i) the trailing 22-day realized volatility of 0050.TW is below 20% annualized ($\mathrm{RV}_{22}^{\mathrm{TW}} < 0.20$), ensuring that local market conditions are genuinely calm; and (ii) the current VIX level is below its trailing 252-day rolling 30th percentile ($\mathrm{VIX}_t < \mathrm{VIX}_t^{p30}$), capturing periods of unusually low U.S. implied volatility *relative to recent history* rather than in absolute terms. The percentile-based VIX filter adapts automatically to changing volatility regimes, avoiding the regime-dependence problem inherent in fixed thresholds.

This hybrid strategy improves the Sharpe ratio by +0.162 over the base 8.63/VIX strategy with zero maximum drawdown penalty, passing stringent statistical validation: the Harvey t-statistic is 4.79 (well above the $t > 3.0$ threshold of Harvey et al. 2016), all 18 out of 18 cross-validated out-of-sample periods produce positive improvements, and 100% of 10,000 bootstrap replications confirm a positive Sharpe increment. Robustness is confirmed on the 0056.TW high-dividend ETF ($t = 5.67$), which excludes TSMC and has a different sector composition, ruling out single-stock dependence.

4.5 Transaction Cost Sensitivity

Taiwan's Securities Transaction Tax differentiates between common stocks (0.30% on sell) and ETFs (0.10% on sell), and online brokerages typically charge 30% of the statutory 0.1425% commission rate. Our baseline applies 0.186% round-trip—the appropriate rate for 0050.TW as an ETF—but a referee may legitimately ask whether the volatility-targeting (VT) advantage survives across the full Taiwan tax-and-commission range. Table 6 reports Sharpe ratios and maximum-drawdown (MDD) improvements over Buy & Hold at four round-trip transaction rates spanning 0.10% to 0.30%. The Sharpe ratio of each strategy declines monotonically with the transaction rate, as expected, but two findings emerge clearly. First, the MDD improvement relative to Buy & Hold survives the entire range: at the worst case of 0.30%, all four VT specifications retain MDD improvements of at least 11.5 percentage points. Second, the monthly-rebalanced 8.63/VIX specification is far less sensitive to transaction costs than the daily-rebalanced VT variants—its Sharpe range across the 0.10%–0.30% grid is only 0.024 versus 0.098–0.127 for the daily-rebalanced strategies—reinforcing our practical recommendation that retail investors should adopt monthly rebalancing.

Table 6: Transaction Cost Sensitivity for VT Strategies on 0050.TW

Strategy	Net Sharpe at round-trip transaction rate				MDD impr. vs BH
	0.10%	0.15%	0.186%	0.30%	at TX=0.30% (pp)
EWMA VT (10%)	0.564	0.540	0.522	0.467	+13.6
GARCH VT (10%)	0.848	0.817	0.795	0.725	+12.3
GJR VT (10%)	0.955	0.923	0.900	0.828	+11.5
8.63/VIX (monthly)	0.967	0.961	0.956	0.943	+21.2

Notes: 0.10% is the ETF-only floor (tax 0.10%, no commission); 0.15% is the midpoint of the Taiwan transaction-tax range; 0.186% is the paper canonical (ETF tax 0.10% + online-discount commission 0.04275% $\times$ 2); 0.30% is the hypothetical common-stock rate (not actually applicable to the 0050.TW ETF; included for upper-bound robustness only). Evaluation windows: EWMA 2010–2026 (n = 3,980); GARCH/GJR 2020–2026 (n = 1,524); 8.63/VIX 2016–2026 (n = 2,497). Sample annual turnovers, identical across TX rates by construction: EWMA 489%/yr, GARCH 634%/yr, GJR 662%/yr, 8.63/VIX 104%/yr. MDD improvement is in percentage points and is computed within each strategy's own evaluation window. Reproduction: `experiments/paper2_R1_transaction_tax_fix/turnover_and_tax.py` using the pinned snapshot `paper/taiwan-vt/data/0050_tw_twii_2330_tw_2317_tw_2454_tw_0056_tw_spy_vix_2008-2026.csv`.

4.6 Linearity Robustness of the VIXTWN-to-VIX Ratio

The calibration $K = 12/1.39 = 8.63$ used in the VIX-proxy VT strategy assumes the VIXTWN-to-VIX ratio is approximately stable across volatility regimes. A referee may legitimately ask whether this linearity assumption breaks in tail-volatility episodes, in which case K would systematically misallocate during the precise periods VT strategies aim to manage. Table 7 reports the ratio conditional on expanding-window VIX quantile buckets (no full-sample lookahead) and on a tail bucket defined as $|\Delta \log \mathrm{VIX}_t| > 2\,\sigma_{t-1}$, using the TAIFEX-official VIXTWN series (K1098 parse, $n = 3{,}075$ post-warmup, 2008–2021).

The ratio declines monotonically from 1.018 at low VIX (Q1) to 0.824 at high VIX (Q4), a 19% relative drop; the tail bucket averages 0.877. Each bucket's 95% stationary-block bootstrap confidence interval ($B = 500$, mean block = 21 days, seed = 42) excludes 1.39. A Kolmogorov-Smirnov two-sample test rejects equality of the tail and non-tail ratio distributions ($D = 0.274$, $p < 0.001$). The static-ratio assumption therefore does not hold strictly, but two mitigating points preserve the substantive conclusions of Section 4. First, the Sharpe ratio of the 8.63/VIX strategy is invariant to K (Section 4, line ∼313); K affects only the risk level and hence maximum drawdown. Second, the recent (Dec 2025–Apr 2026, $n = 76$) ratio of 1.393 used in the calibration reflects a structurally higher-volatility regime in Taiwan; our recommendation is to interpret $K = 8.63$ as a recent-window practical calibration and to disclose the regime dependence rather than to revise the calibration.

Table 7: VIXTWN-to-VIX Ratio Conditional on VIX Regime, 2008–2021

Bucket	n	Mean	Median	SD	95% CI	Contains 1.39?
Q1 (VIX $\leq$ Q25)	1,533	1.018	1.018	0.141	$[0.987, 1.046]$	No
Q2 (Q25–Q50)	634	1.007	0.990	0.180	$[0.955, 1.054]$	No
Q3 (Q50–Q75)	574	0.947	0.919	0.176	$[0.897, 0.995]$	No
Q4 (VIX $>$ Q75)	334	0.824	0.824	0.133	$[0.785, 0.862]$	No
Tail (2σ)	183	0.877	0.877	0.202	$[0.832, 0.919]$	No
Overall (long sample)	3,075	0.981	0.985	0.180	$[0.953, 1.009]$	No
Recent (Dec 2025–Apr 2026)	76	1.393	–	–	—	—

Notes: Quantile thresholds are computed in an expanding-window manner using observations strictly prior to day t; the first 252 days are dropped as warmup. The tail bucket uses $|\Delta \log \mathrm{VIX}_t| > 2\,\sigma_{t-1}$ where σ_{t-1} is the expanding standard deviation of log-VIX changes through $t-1$. Confidence intervals are computed via stationary block bootstrap [?] with $B = 500$ replications and mean block length 21 trading days. The long-history sample uses the TAIFEX-official VIXTWN series; the recent window uses K1181's daily snapshot. The decline in the ratio at high VIX reflects under-reaction of VIXTWN to U.S. vol shocks rather than over-reaction in calm periods; mechanistically this is consistent with structural-break evidence around the 2020 launch of an active VIXTWN futures market.

5 Macroeconomic Indicators for Taiwan Volatility

While our three core contributions focus on leverage amplification, VT strategies, and time-zone arbitrage, we briefly examine whether Taiwan-specific macroeconomic indicators provide incremental volatility forecasting power beyond VIX. An important question for practitioners is whether Taiwan-specific macroeconomic variables can enhance volatility forecasts or VT strategies beyond what VIX alone provides. We conduct a comprehensive sweep of 27 Taiwanese macroeconomic indicators within a GARCH-MIDAS framework [Engle et al., 2013], testing each variable's incremental contribution to 0050.TW volatility forecasting.

5.1 Import Growth as a Volatility Predictor

Among the 27 indicators tested, only one achieves out-of-sample significance: year-over-year import growth (partial $r = 0.214$, OOS improvement +5.6% over base GJR-GARCH, Diebold-Mariano $p = 0.043$). Taiwan is a highly trade-dependent economy, with imports closely tracking global semiconductor demand and industrial production. When import growth exceeds 20%—indicating overheated external demand—adding a "macro guard" that reduces VT exposure improves the Sharpe ratio by 0.056 but worsens maximum drawdown by 3.2 percentage points, an unfavorable tradeoff.

5.2 Null Results

The following indicators fail to improve out-of-sample volatility forecasts or VT performance after controlling for VIX:

- **Business cycle indicator levels**: The National Development Council's composite business cycle score ("traffic light" system) has no predictive power for next-month realized volatility ($t = -0.53$, $p = 0.60$), consistent with its well-documented lagging nature. Industrial production and M2 money supply growth likewise show insignificant incremental explanatory power (partial $r < 0.05$). However, the *directional change* in these indicators tells a different story; see Section 5.3 below.

- **Margin trading**: Net margin balance and short interest do not predict future volatility or returns. Margin trading in Taiwan appears to be a coincident or lagging indicator rather than a leading indicator.

- **Foreign institutional flows**: Net foreign buying/selling of Taiwanese equities is a lagging indicator, reacting to price movements rather than preceding them.

- **Put/call ratio**: The Taiwan options market put/call ratio shows a contemporaneous correlation of $r = 0.41$ with same-day volatility, but the lagged version drops to zero, confirming that it contains no predictive information.

- **TSMC valuation**: As the dominant constituent of 0050.TW, TSMC's price-earnings ratio might be expected to predict volatility. It does not: the relationship is driven by the AI-related structural break in TSMC's earnings, which invalidated historical PE-volatility patterns.

These null results are consistent with our broader finding that VIX serves as a "sufficient statistic" for volatility-related information in markets with strong U.S. linkages. For U.S. equities, we have documented that VIX absorbs the predictive content of more than ten alternative indicators (credit spreads, yield curves, VVIX, MOVE, CNN Fear & Greed Index, AAII sentiment, VIX term structure). For Taiwan, VIX is not perfectly sufficient—import growth and business cycle momentum provide incremental value—but the gap is small enough that the practical benefit of monitoring additional indicators is modest for most investors.

5.3 Business Cycle Indicator Momentum

Beyond trade data, we examine Taiwan's composite business cycle indicators published by the National Development Council. As noted above, the business cycle score itself has no predictive power for next-month realized volatility ($t = -0.53$, $p = 0.60$), consistent with its well-documented lagging nature. However, the month-over-month change in the leading indicator composite achieves the strongest predictive signal among all Taiwan-specific variables ($t = 3.74$, $p < 0.001$, $R^2 = 7.1\%$), surpassing even trade data. A coincident indicator momentum strategy—reducing equity exposure when the coincident composite declines for three consecutive months—yields Sharpe 0.732 (OOS 2018–2024: 1.260), substantially exceeding the 8.63/VIX baseline of 0.334 over the same 2018–2024 sub-period (compared to 1.137 over the full 2016–2026 sample; see Table 11).[5] The key insight is that the business cycle indicator's value lies in its *directional change*, not its level.

This finding has broader methodological implications: macroeconomic indicators that appear uninformative in level form may contain substantial predictive content in their first differences. The leading indicator momentum captures regime shifts in the Taiwanese economy—transitions between expansion and contraction—that are orthogonal to the high-frequency volatility information embedded in VIX.

5.4 Ex-Dividend Volatility and VT Implications

Taiwan's concentrated ETF dividend calendar—0050.TW and 0056.TW typically go ex-dividend within the same two-week window each July—creates a predictable volatility spike. We examine realized volatility in the five trading days following ex-dividend dates and find an increase of +32% for 0050.TW and +69% for the high-dividend 0056.TW relative to the 20-day pre-event baseline. The larger effect for 0056.TW is consistent with its higher dividend yield (∼5–7% versus ∼3% for 0050.TW) and its retail-investor-heavy shareholder base, which generates more post-ex-date selling pressure.

Despite this pronounced volatility effect, the historical fill rate—the proportion of ex-dividend gaps that are fully recovered within 60 trading days—is 79% for 0050.TW and 90% for 0056.TW over the 2012–2025 sample. This high fill rate implies that the ex-dividend volatility spike is transient and mean-reverting, driven by mechanical price adjustment and short-term tax-motivated trading rather than fundamental information.

For VT practitioners, a practical question is whether the ex-dividend volatility spike requires special treatment (e.g., temporarily reducing the target volatility or applying a

[5] The difference between sub-period and full-sample Sharpe ratios for the 8.63/VIX strategy reflects the strategy's underperformance during the prolonged low-volatility bull market of 2020–2021, when the VIX-scaled weight held substantially less than full exposure.

dividend-adjustment filter). We find that it does not: because VT strategies mechanically reduce exposure when realized volatility rises, the EWMA and GARCH models automatically scale down the position during the post-ex-dividend period without any manual intervention. The ex-dividend effect is therefore self-correcting within the VT framework, and no dividend-specific adjustment is necessary.

6 Earnings Announcement Volatility: A Universal Cross-Market Regularity

Beyond the aggregate volatility dynamics examined in previous sections, our pooled panel estimation reveals a robust firm-level empirical regularity: a universal surge in stock-level volatility on earnings announcement days that is positive, statistically significant, and consistent in magnitude ordering across three geographically and institutionally distinct equity markets. This finding emerges from the A4f-EAV model, a pooled panel GARCH specification augmenting the standard volatility equation with an earnings announcement indicator.

6.1 A4f-EAV Model Specification

The A4f-EAV specification models the conditional variance of firm i on day t as:

$$\sigma_{i,t}^2 = \omega_i + \alpha\varepsilon_{i,t-1}^2 + \beta\sigma_{i,t-1}^2 + \theta_{\mathrm{VIX}} \cdot \mathrm{VIX}_t + \theta_{\mathrm{EAV}} \cdot \mathbb{1}[\mathrm{EA}_{i,t}], \tag{7}$$

where $\mathbb{1}[\mathrm{EA}_{i,t}]$ equals one on firm i's quarterly earnings announcement day and zero otherwise. Firm fixed effects ω_i are absorbed by profile likelihood; $(\alpha, \beta, \theta_{\mathrm{VIX}}, \theta_{\mathrm{EAV}})$ are pooled across all firms in the panel. We estimate via maximum likelihood over the full pooled sample.

The parameter of interest is θ_{EAV}: conditional on the VIX level and lagged GARCH dynamics, a positive estimate implies that earnings announcements carry incremental volatility orthogonal to systematic risk and variance momentum. VIX is included as a time-varying volatility floor that absorbs broad market uncertainty common across firms, ensuring θ_{EAV} reflects firm-specific announcement risk rather than market-wide elevated uncertainty that may coincide with heavy reporting calendars.

6.2 Three-Market Evidence

Table 8 reports the primary estimates for Taiwan (N=31, 2010–2025), the United States (N=30 S&P 500 large-caps, 2014–2025), and Japan (N=30 TOPIX large-caps, 2014–2025). In all three markets, $\hat{\theta}_{\text{EAV}}$ is strictly positive and highly significant under both Hessian-based and cluster bootstrap standard errors.

Table 8: Earnings Announcement Volatility Across Three Markets (A4f-EAV)

	Taiwan (K1145)	United States (K1147)	Japan (K1150)
$\hat{\theta}_{\text{EAV}}$	6.36×10^{-5}	1.91×10^{-4}	1.41×10^{-4}
t_{Hessian}	14.14^{***}	22.39^{***}	20.16^{***}
$t_{\text{bootstrap}}$	5.24^{***}	4.50^{***}	11.99^{***}
95% CI (bootstrap)	$[4.13,\ 9.38] \times 10^{-5}$	$[1.29,\ 2.80] \times 10^{-4}$	$[1.29,\ 1.76] \times 10^{-4}$
Stocks (N)	31	30	30
Pooled obs.	121,014	90,479	87,917
Sample period	2010–2025	2014–2025	2014–2025

Notes: Profile-likelihood pooled panel MLE. Hessian t-statistics use analytic (inverse-Hessian) standard errors; bootstrap estimates and t-statistics use stock-level block bootstrap ($B = 150$, seed 42). 95% CIs are bias-corrected bootstrap percentile intervals. *** denotes $p < 0.001$ under Hessian inference. Source: `main_fit_eav_window_1` + `cluster_bootstrap` in K1145 (TW), K1147 (US), K1150 (JP) results JSON.

The magnitude ordering is US > JP > TW: $\hat{\theta}_{\text{EAV}}$ is approximately three times larger in the United States and 2.2 times larger in Japan than in Taiwan. This cross-market ordering is economically interpretable. U.S. quarterly earnings are subject to the highest analyst coverage, the most liquid options markets, and the shortest information halflife once announcements are released; these institutional features amplify the realized-volatility response. Japan's intermediate position likely reflects its high degree of institutional participation, synchronized reporting requirements under Tokyo Stock Exchange rules, and active arbitrage by domestic funds around announcement dates. Taiwan's relatively lower coefficient is consistent with the dominance of export-oriented semiconductor constituents—firms whose earnings surprises are partly anticipated from global semiconductor cycle data (equipment orders, memory pricing) that circulates before formal announcements, attenuating the incremental information release. Importantly, the direction of the effect is identical across all three markets: earnings announcements uniformly elevate volatility beyond what VIX and GARCH dynamics would predict.

6.3 Five-Layer Robustness

Table 9 reports robustness across five layers: inference method, event window definition, and sample composition. The earnings announcement volatility effect is not an artifact of any single modelling choice.

Table 9: Five-Layer Robustness of $\hat{\theta}_{\mathrm{EAV}}$ Across Specification Variants

Layer	Specification	TW ($\hat{\theta}$)	US ($\hat{\theta}$)	JP
1. Hessian SE	Baseline window-1, analytic	6.36×10^{-5}	1.91×10^{-4}	$1.41 \times$
2. Bootstrap SE	Baseline window-1, $B = 150$	6.77×10^{-5}	1.95×10^{-4}	$1.49 \times$
3. Window ± 1	Three-day event window	3.80×10^{-5}	7.73×10^{-5}	$1.10 \times$
4. Window ± 2	Five-day event window	1.73×10^{-5}	8.29×10^{-5}	$8.12 \times$
5. Dropout	20% stocks omitted (5 seeds)	$[6.21,\ 7.96] \times 10^{-5}$	$[1.82,\ 2.15] \times 10^{-4}$	$[1.34,\ 1.4$

Notes: Window $\pm k$ means the event indicator spans $2k + 1$ days centred on the announcement date; the per-day $\hat{\theta}_{\mathrm{EAV}}$ declines as the window widens because the concentrated announcement-day effect is spread across a larger denominator. Dropout reports the range of $\hat{\theta}_{\mathrm{EAV}}$ over five 20%-subsample draws (seed 42). All 15 cells are strictly positive; market ordering US > JP > TW is preserved across all five layers. Source: `robustness_eav_window`, `cluster_bootstrap`, `robustness_dropout` in K1145 (TW), K1147 (US), K1150 (JP).

Three patterns emerge from Table 9. First, $\hat{\theta}_{\mathrm{EAV}}$ is strictly positive across all 15 specification-market combinations; the effect is sign-stable. Second, bootstrap and Hessian point estimates are close in every market, implying the analytic standard errors underestimate uncertainty (as expected in panel models with within-stock temporal dependence) but do not materially bias the point estimate itself. Third, the dropout ranges are narrow relative to the baseline estimates—TW spans 28%, US and JP span approximately 17%—confirming that no small subset of firms drives the regularity. See Section 8.8 (in the Discussion) for the formal Bonferroni correction and Wald test addressing multiple-testing concerns.

7 VaR and Risk Management

7.1 VaR Methodology

We compute one-day Value-at-Risk at the 1% level using GJR-GARCH conditional volatility combined with the Student-t distribution (degrees of freedom = 5):

$$\mathrm{VaR}_{1\%,t} = \hat{\sigma}_t \cdot q_{0.01}^{t(5)} \tag{8}$$

where $q_{0.01}^{t(5)} = -3.365$ is the 1% quantile of the standardized Student-t distribution with 5 degrees of freedom.

We also evaluate the skewed Student-t distribution, which simultaneously estimates degrees of freedom (η) and skewness (λ) via maximum likelihood, adapting to each asset's specific tail behavior. For 0050.TW, the estimated parameters are $\eta = 5.2$ and $\lambda = -0.05$ (near-symmetric with moderate fat tails).

7.2 Backtest Results

Over the out-of-sample period 2020–2026 (1,501 trading days), the GJR-GARCH + Student-t(5) VaR produces 8 violations (0.5%), within the Basel III Green Zone threshold of 1.0% (noting that the traffic-light framework is defined for 250-day windows; we apply the 1% threshold as a benchmark across our full sample). The Kupiec [1995] likelihood ratio test does not reject correct unconditional coverage ($p = 0.12$). For comparison, the Normal distribution produces 30 violations (2.0%), a borderline result that would trigger the Yellow Zone under Basel III.

The EWMA + Student-$t(5)$ combination performs comparably, with 8 violations (0.5%), demonstrating that the simpler EWMA model is sufficient for regulatory VaR compliance in the Taiwanese market. This is a practical finding: Taiwan investors can implement a fully compliant VaR framework using only EWMA (no GARCH optimization required) plus the Student-t critical value.

7.3 Skewed Student-t for Taiwan

The skewed Student-t distribution passes the Kupiec [1995] test for all six assets in our broader cross-asset panel (SPY, QQQ, GLD, TLT, EEM, and 0050.TW)—the only distribution to achieve a perfect 6/6 pass rate. The Cornish-Fisher expansion achieves 5/6 (failing for QQQ due to excessive conservatism), while the fixed Student-$t(5)$ achieves 4/6 and the Normal distribution achieves only 1/6. For Taiwan specifically, both skewed Student-t and fixed Student-$t(5)$ perform well, reflecting the near-symmetric conditional return distribution of 0050.TW.

7.4 Cornish-Fisher Divergence

An important caveat for the Taiwanese market is that the Cornish-Fisher (CF) VaR expansion diverges when using a rolling window of $w = 2000$ for 0050.TW. The issue is that kurtosis estimates become extremely large (up to 545 in some windows) due to occasional extreme returns that are not washed out by the long window. With $w = 504$, CF-VaR performs well ($p = 0.88$), but the shorter window introduces persistence bias in the GARCH estimates. Winsorization of extreme kurtosis values resolves the divergence, but this adds an ad hoc element that undermines the method's theoretical appeal. For practical purposes, we recommend the skewed Student-t approach for Taiwan VaR, which is both theoretically principled and computationally stable.

7.5 Expected Shortfall Analysis

Basel III (2016–2019) mandates Expected Shortfall (ES) as the primary tail-risk metric for internal model approval, superseding VaR. We compute ES backtests over 2019–2026 (n = 1,756 trading days) using the Acerbi and Szekely [2014] Z_2 statistic, which tests whether the conditional mean of excess losses equals the ES forecast, with p-values from 1,000 bootstrap replications. We also report the Fissler and Ziegel [2016] consistent joint VaR+ES scoring function (lower is better), which provides a coherent ranking of tail forecast accuracy.

Table 10 reports results at the 1% level. A key finding is that *all four distributional specifications pass the Acerbi-Székely ES test*, including GJR+Normal, which fails both the Kupiec VaR test and the Basel III Trinity. This pattern reflects a distributional complementarity: VaR tests quantile coverage (frequency of exceedance), while ES tests conditional mean severity given exceedance. A model miscalibrated in quantile frequency can remain well-calibrated in the expected magnitude of tail losses.

Table 10: Expected Shortfall Backtest: 0050.TW, 2019–2026

Model	Mean VaR (%)	Mean ES (%)	ES/VaR ratio	Z_2 stat (p-value)	FZ score
GJR + Normal	2.71	3.11	1.15	3.01 (0.52)	4.055
GJR + Student-t	3.03	4.00	1.32	2.07 (0.55)	4.025
GJR + Hist. Sim.	3.04	4.02	1.32	2.08 (0.46)	4.011
GJR + Cornish-Fisher	3.83	5.33	1.39	1.49 (0.54)	3.913

Notes: OOS period 2019-01-01 to 2026-04-02 (n = 1,756 days). Mean VaR and Mean ES at the 1% level expressed as percentage of daily portfolio value. Z_2: Acerbi and Szekely [2014] expected shortfall test; all models fail to reject the null of correct ES calibration at the 5% level. FZ score: Fissler and Ziegel [2016] consistent joint VaR+ES scoring function (lower = better). Source: `k896_taiwan_es_supplement_results.json`.

The Cornish-Fisher specification achieves the lowest Fissler-Ziegel score (3.91), indicating the highest joint accuracy in estimating tail severity, despite failing the Kupiec VaR test due to over-conservatism (9 violations, 0.5%, below the 1% target).GJR+Student-t remains the recommended specification for practical use: it achieves VaR Trinity pass, ES acceptance ($Z_2 = 2.07$, $p = 0.55$), and a mean ES of 4.00%, providing a well-calibrated upper bound on expected tail losses for Taiwan investors.

8 Discussion

8.1 Currency Risk

An important practical consideration for Taiwanese investors considering U.S.-linked strategies is currency risk. The TWD/USD exchange rate introduces additional volatility that reduces the Sharpe ratio of SPY-denominated strategies by approximately 18% when converted to TWD terms. However, the USD tends to appreciate during equity market crises (a safe-haven effect), providing a partial natural hedge: during the 2022 drawdown, USD appreciation added approximately 15% to returns measured in TWD.

The 8.63/VIX strategy on 0050.TW (Sharpe 1.14 in TWD, no currency risk) outperforms the 12/VIX strategy on SPY converted to TWD (Sharpe 0.67 including FX effects). Given the additional complexity and cost of currency hedging (estimated at approximately 2% per annum for TWD/USD forwards), we do not recommend FX hedging for Taiwanese investors implementing VIX-based strategies on local equities.

8.2 Practical Implications

Our results suggest several practical implications for individual investors in Taiwan, ordered by implementation complexity:

VIX Step Rule. The simplest approach requires no computation: if VIX < 15, allocate 100% to 0050.TW; if VIX $\in [15, 25]$, allocate 70%; if VIX > 25, allocate 40%. This captures the bulk of VT's drawdown protection with monthly rebalancing.

8.63/**VIX strategy.** Compute $w = 8.63/\text{VIX}$ at each month-end, allocating w% to 0050.TW and $(1 - w)$% to a short-term bond fund. Transaction cost: approximately 0.15% per rebalancing.

Insurance cost quantification. A key practical question is: how much does VT's drawdown protection cost in terms of foregone returns? For 0050.TW, the EWMA VT strategy reduces maximum drawdown from -77.3% to -48.6%—a benefit of 28.8 percentage points, approximately double the U.S. drawdown reduction—but at a CAGR cost of 3.88 percentage points (from 7.59% to 3.71%). This translates to a cost of approximately

13.5 basis points of annual return per percentage point of MDD improvement. For comparison, the analogous cost for SPY in the U.S. is substantially lower (approximately 5–8 bp per percentage point of MDD improvement), reflecting the higher opportunity cost of defensive positioning in a market with greater upside volatility. Taiwan VT therefore provides *more* insurance (larger MDD improvement) but at a *higher* price (greater CAGR sacrifice). This cost–benefit tradeoff is particularly relevant for risk-averse investors with CRRA utility $\gamma \geq 5$, who value the MDD reduction more than the CAGR loss, consistent with the utility analysis in Moreira and Muir [2017].

8.3 Comparison to Prior Literature

Our findings both confirm and extend the existing VT literature. Consistent with Moreira and Muir [2017], VT's primary benefit is drawdown reduction rather than Sharpe ratio improvement; the Sharpe improvement t-statistic is only 0.33, far below conventional significance levels, but the MDD improvement is statistically significant (bootstrap $p = 0.0004$). This result holds in both U.S. and Taiwanese markets, suggesting it is a universal property of VT strategies.

The diversification amplification of the leverage effect extends the work of Ang and Chen [2002] on asymmetric correlations. While Ang and Chen document that correlations increase during downturns, we show that this phenomenon mechanically amplifies the GJR-GARCH γ parameter at the index level relative to the constituent level, and that the amplification ratio varies across markets (approximately 4.3× for the broad TAIEX based on $N = 9$ stocks with 90% CI [2.28, 6.58]; canonical BW-robust specification, K1370; 1.8× for the investable 0050.TW under rolling-window specification, versus 2.8× for the U.S.). The gap between TAIEX and 0050.TW ratios highlights the importance of distinguishing between broad market indices and investable vehicles when quantifying the leverage effect.

Supplementary evidence on cross-market information transmission (Appendix A) relates to the broader literature on international return predictability [Eun and Shim, 1989, Rapach et al., 2013]. The finding that approximately 78% of the close-to-close alpha is absorbed by the overnight opening gap—before any trader can act—provides a quantitative measure of price discovery speed across time zones and motivates the use of VIX as a cross-market volatility proxy for Taiwan VT.

8.4 Combining Macro Signals with VIX-Based Strategies

A natural question is whether Taiwan-specific macro signals can enhance the VIX-based framework. We find that combining VIX scaling with the leading indicator direction—using $k = 10$ (aggressive) when the leading indicator is rising and $k = 6$ (conservative) when declining—significantly improves the pure 8.63/VIX strategy over the full 2012–2026 sample (Diebold-Mariano $p = 0.0005$, Sharpe from 0.999 to 1.151; see Table 11). The intuition is straightforward: when the leading indicator signals economic expansion, the VT strategy can afford to take more risk; when contraction looms, it should be more defensive. This conditional approach effectively introduces regime awareness into an otherwise purely volatility-driven framework.

However, this monthly-frequency macro signal cannot improve the daily-frequency SPY momentum strategy (DM $p = 0.82$; see Appendix A for the time-zone momentum analysis), confirming that effective signal combination requires matching timescales. The SPY momentum signal already captures high-frequency regime information through daily U.S. market movements, leaving no room for a slower-moving macro indicator to add value. This timescale-matching constraint has practical implications for investors using the monthly VIX-rebalancing strategy, who can benefit meaningfully from incorporating leading indicator direction.

8.5 TSMC Concentration Robustness

A natural concern is whether our VT results for 0050.TW are driven entirely by its dominant constituent, TSMC, which accounts for approximately 50% of the index weight and has seen its rolling beta to 0050.TW rise from 0.38 in 2009 to 0.72 by 2026. We address this concern with a decomposition analysis.

We apply VT separately to TSMC (2330.TW) and to a synthetic ex-TSMC 0050.TW portfolio (constructed by removing TSMC's estimated contribution from daily 0050.TW returns). TSMC VT achieves a Sharpe ratio of 1.121, exceeding the 0050.TW VT Sharpe of 0.936. Importantly, the ex-TSMC VT portfolio also produces positive risk-adjusted returns (Sharpe ranging from 0.193 to 0.637 depending on assumptions about TSMC's time-varying weight), confirming that VT benefits are not solely attributable to TSMC.

An unexpected finding from this decomposition concerns the leverage effect. The 0050.TW index exhibits a GJR-GARCH leverage parameter of $\gamma = 0.124$ ($t = 2.46$), whereas TSMC alone shows $\gamma = 0.054$ ($t = 1.07$, insignificant). This result—stronger leverage in the diversified index than in its dominant constituent—reinforces the diversification amplification mechanism documented in Section 3: aggregation across stocks "cleans" the systematic leverage signal by averaging out firm-specific idiosyncratic noise, leaving a purer asymmetric volatility response. TSMC's relatively weak individual leverage effect is consistent with its status as a growth-oriented technology stock whose volatility dynamics are driven more by earnings and semiconductor cycle surprises than by the classical leverage channel of Black [1976].

We acknowledge the growing concentration risk: TSMC explains 52.5% of 0050.TW return variance over the full sample, and its rolling beta has approximately doubled over the sample period. As TSMC's weight continues to increase, the distinction between 0050.TW VT and single-stock TSMC VT will narrow. Investors concerned about concentration may consider the 0056.TW ETF (which excludes TSMC) as an alternative VT substrate, though at the cost of weaker VIX-based signal quality due to the lower U.S. technology exposure.

8.6 Cross-Market Validation of U.S. VT Findings

A natural question is whether the key findings documented for U.S. VT strategies generalize to Taiwan, or whether market-specific conditions invalidate their applicability. We conduct a systematic cross-validation of four core U.S. results using 0050.TW data over the matched 2008–2026 sample.

VIX sufficiency generalizes. The U.S. finding that VIX serves as a "sufficient statistic” for volatility forecasting extends to Taiwan: VIX alone explains $R^2 = 0.257$ of next-day 0050.TW realized volatility variance. Adding 0050.TW’s own lagged realized volatility raises R^2 to 0.387 (a gain of +0.130), indicating that own-RV provides more incremental information for Taiwan than for U.S. equities, but the VIX-only model already captures the majority of predictable variation. This result is consistent with the SSVS evidence in Section 3.3 and supports the practical viability of VIX-only VT strategies in markets with strong U.S. linkages.

Optimal allocation partially generalizes. The U.S. finding that a 50/50 equity/gold allocation is approximately optimal shifts but does not fully generalize to Taiwan. A grid search over 0050.TW/SPY allocations reveals an optimal mix of approximately 40% 0050.TW and 60% SPY, reflecting the moderately higher volatility of the Taiwanese ETF (annualized vol $\approx$ 18%) relative to SPY ($\approx$ 16%). The allocation is closer to vol-parity than previously thought, as Taiwan’s true baseline volatility—once corrected for historical data artifacts—is only marginally above SPY. For three-asset portfolios including gold, the optimal mix is approximately 40% 0050.TW, 30% SPY, and 30% GLD. These results imply that Taiwanese investors can maintain a meaningful home-bias component while still capturing U.S. equity returns and gold diversification benefits.

Calendar anomalies generalize (both null). Neither U.S. nor Taiwan markets exhibit statistically significant calendar-based volatility anomalies (day-of-week, turn-of-month) that survive the Harvey et al. [2016] threshold, confirming that calendar effects are not a reliable basis for VT strategy design in either market.

Optimal rebalancing frequency does not generalize (preliminary). U.S. VT strategies are relatively insensitive to rebalancing frequency (monthly is near-optimal). For 0050.TW, preliminary results suggest that daily rebalancing may be preferred, potentially due to the elevated leverage amplification creating faster volatility mean-reversion. However, this result should be treated as preliminary: an independent review identified a potential holiday-calendar alignment issue in the cross-market daily return matching, which could bias the rebalancing frequency comparison.[6]

In summary, three of four core U.S. findings generalize or partially generalize to Taiwan (VIX sufficiency, optimal allocation closer to vol-parity, calendar null results), while one does not (rebalancing frequency). The allocation result—40/60 rather than 50/50—is economically intuitive given Taiwan's moderately higher baseline volatility (≈18% vs. ≈16% for SPY). The rebalancing frequency difference reflects Taiwan's structural characteristics, including stronger leverage amplification and the cross-market information lag.

8.7 Sharpe Ratio Reconciliation

Table 11 reconciles the Sharpe ratios reported throughout the paper for the 8.63/VIX strategy and its variants. The apparent discrepancies arise from differences in sample period, rebalancing frequency, and strategy specification (pure VIX scaling versus the VIX + leading indicator combination).

[6]Specifically, Taiwan and U.S. markets have different holiday calendars, creating occasional mismatches in daily return alignment. While we align on trading dates, the interaction between non-overlapping holidays and the lagged VIX signal may introduce spurious daily-frequency effects. We are conducting additional robustness checks with explicit holiday-calendar harmonization.

Table 11: Sharpe Ratio Reconciliation for VIX-Based Strategies on 0050.TW

Strategy	Period	Sharpe	Table/Section
8.63/VIX (monthly, lagged)	2016–2026 (full)	1.137	Table 4
8.63/VIX (monthly, lagged)	2018–2024 (sub-period)	0.334	Section 5.3
8.63/VIX + Leading indicator	2012–2026 (full)	0.999	Section 8
VIX + Leading indicator combo	2012–2026 (full)	1.151	Section 8

Notes: All strategies use monthly rebalancing with lagged weights (VIX_{t-1} determines the weight for period t) and include transaction costs of 0.186% round-trip. The 2018–2024 sub-period Sharpe is lower because this window captures the prolonged low-volatility bull market of 2020–2021, during which the VIX-scaled strategy was underinvested relative to buy-and-hold. The VIX + Leading indicator combination uses $k = 10$ when the leading indicator rises and $k = 6$ when it declines; its longer sample (2012–2026) reflects the availability of leading indicator data from 2012.

8.8 Self-Challenge: Conservative Inference for the Universal EAV Regularity

A natural concern about the EAV evidence in Section 6 is multiple-testing inflation: testing the same null hypothesis across three markets creates three opportunities for spurious rejection. We apply the Bonferroni family-wise error rate (FWER) correction with $k = 3$ simultaneous tests, which requires each individual p-value to fall below $0.05/3 = 0.0167$ rather than the conventional 0.05. Under a normal approximation to the cluster bootstrap distribution, this corresponds to a one-sided critical value of $z^* \approx 2.39$. The three bootstrap t-statistics are 5.24 (TW), 4.50 (US), and 11.99 (JP)—all exceed the Bonferroni threshold by comfortable margins. The FWER-adjusted conclusion is that earnings announcement volatility is statistically distinguishable from zero in every market individually, even after correcting for testing across three markets simultaneously.

A second challenge concerns the large discrepancy between Hessian and cluster bootstrap t-statistics. The compression is roughly 3× for TW ($14.14 \to 5.24$) and 5× for US

(22.39 $\to$ 4.50). This is expected in pooled panel models when residuals exhibit within-stock temporal dependence across announcement cycles and when the number of clusters (N = 30–31 stocks) is moderate. The Hessian treats all $\sim$90,000–121,000 observations as independently drawn, underweighting the effective degrees of freedom; the block bootstrap accounts for within-stock serial dependence by resampling entire stock-time blocks. We therefore designate the cluster bootstrap t-statistic as the primary inferential basis in Section 6; the Hessian t-statistics are reported as a computational benchmark, not as the preferred inferential standard.

For completeness, the Wald statistic for the joint hypothesis $H_0 : \theta_{\text{EAV,TW}} = \theta_{\text{EAV,US}} = \theta_{\text{EAV,JP}} = 0$ under the Hessian covariance (treating markets as independent) is $\chi^2(3) >$ 1,000 (each individual Hessian t exceeds 14), yielding $p < 10^{-200}$. While this extreme value reflects the Hessian's downward bias in standard errors, even under the bootstrap standard errors the evidence is decisive: the weakest of the three bootstrap t-statistics is 4.50 (US), which corresponds to $p \approx 3 \times 10^{-6}$ under a standard normal approximation, clearing the Bonferroni-adjusted threshold by four orders of magnitude. The universal EAV regularity is not a borderline finding that depends on inference assumptions; it is robust to the full range of standard error specifications examined.

9 Conclusion

This paper examines volatility targeting strategies in the Taiwan stock market, contributing two sets of findings.

First, we document that the TAIEX exhibits a strong leverage effect (γ = 0.272, t = 3.18, rolling-window specification) that is amplified approximately 4.3× relative to individual stock asymmetries for the broad index under the canonical full-sample BW-robust specification ($\sim$900 stocks; 90% bootstrap CI: [2.28, 6.58], n_{valid} = 1,000 of 1,000 replicates; K1370 v2), while the investable 0050.TW (50 stocks, γ = 0.097) shows a more modest 1.8× amplification under rolling-window specification. The wide confidence interval reflects the small cross-section (N = 9), and a broader sample of

0050.TW constituents would be needed to establish the ratio with greater precision. This amplification arises from higher correlation asymmetry among Taiwanese equities during market declines and has direct implications for model selection: index-level strategies should account for asymmetric volatility even when individual constituents show weak asymmetry.

Second, we evaluate EWMA, GARCH, GJR-GARCH, and VIX-proxy VT strategies adapted for the Taiwan market. Over a common evaluation period (2020–2026), all VT strategies improve both Sharpe ratios and maximum drawdowns relative to buy-and-hold, with GJR VT achieving the highest Sharpe (1.108). Over the longer 2010–2026 sample (K1175 canonical replication), EWMA VT changes the Sharpe ratio from 0.799 (buy-and-hold) to 0.701 while reducing maximum drawdown from -33.8% to -21.2% (a 12.6 percentage point reduction). The VIX-proxy strategy (8.63/VIX), calibrated using the VIXTWN-to-VIX ratio of 1.39, achieves comparable drawdown protection with only monthly rebalancing. A key finding is the disconnect between forecast accuracy and portfolio performance: GJR-GARCH does not significantly outperform GARCH in QLIKE (DM $p = 0.86$), yet GJR-based VT produces a meaningfully higher Sharpe ratio (+0.124), suggesting that tail-specific accuracy matters more than average accuracy for VT applications. We further show that conditional leverage—applying $1.5\times$ allocation when both local realized volatility and VIX percentile are low—improves the Sharpe ratio by +0.162 with zero drawdown penalty ($t = 4.79$, 18/18 cross-OOS periods positive), though this benefit is specific to low-to-moderate volatility assets and should not be applied uniformly. A systematic cross-market validation shows that three of four core U.S. VT findings generalize or partially generalize to Taiwan (VIX sufficiency, allocation near vol-parity, calendar null results), while one does not (rebalancing frequency).

Supplementary evidence on cross-market information transmission (Appendix A) documents a persistent channel from U.S. to Asian equity markets driven by non-overlapping trading hours. Approximately 78% of the close-to-close alpha is absorbed by the overnight opening gap, consistent with efficient price discovery by the opening auction. This channel motivates the use of U.S. VIX as a proxy for Taiwan implied volatility.

Several limitations warrant mention. First, the VIXTWN data history is limited to 64 months (November 2020 to present), which restricts our ability to validate the VIX-to-VIXTWN ratio across different market regimes. As VIXTWN data accumulates, the calibration should be periodically updated. Second, the diversification amplification ratio is estimated from only $N = 9$ individual stocks, yielding wide confidence intervals; expanding to all 50 constituents of 0050.TW would improve precision. Third, our VaR analysis uses a rolling window of 2,000 days, which may be slow to adapt to structural breaks in the Taiwanese market; shorter windows (504 days) improve VaR compliance but introduce GARCH persistence bias. Fourth, we test only a single emerging Asian market in depth; replication in other markets with similar characteristics (e.g., South Korea, where preliminary results are promising) would strengthen the generalizability of our findings. Fifth, TSMC's growing weight in 0050.TW (rolling beta increasing from 0.38 to 0.72 over our sample period) means that our results increasingly reflect the dynamics of a single dominant stock; while the TSMC robustness analysis in Section 8.5 confirms that VT benefits extend to the ex-TSMC portfolio, future studies should monitor whether continued concentration erodes the diversification amplification mechanism.

Future research directions include: (1) validating the VIXTWN-based strategy when sufficient data become available (currently 64 months; 252+ months recommended for comprehensive evaluation); (2) expanding the diversification amplification analysis to a broader cross-section of Taiwanese stocks; and (3) investigating whether the diversification amplification phenomenon has increased over time as passive ETF investing has grown in Taiwan, potentially strengthening the leverage effect at the index level.

A Time-Zone Information Transmission: Supplementary Evidence

This appendix presents supplementary evidence on cross-market information transmission from U.S. to Asian equity markets. While related to the VIX-proxy motivation in the main text (the same cross-market channel that transmits volatility information also transmits return information), the time-zone analysis constitutes a distinct microstructure contribution. We present it here to maintain the main paper's focus on VT strategy evaluation.

A.1 Motivation and Hypothesis

The cross-market spillover documented in Section 3.3 suggests that U.S. market information is impounded into Taiwan prices with a lag. If this lag is systematic and predictable, two questions arise: first, does it create a trading opportunity? Second, how efficiently do Asian markets incorporate this overnight information [Barclay and Hendershott, 2003]? We hypothesize that the sign and magnitude of recent U.S. equity returns predict next-day Taiwan returns, and examine both the implementability of a momentum-based switching strategy and the efficiency of the opening auction as an information aggregation mechanism.

Specifically, we construct a time-zone momentum strategy as follows: if the trailing 10-day cumulative SPY return is positive, hold 0050.TW; otherwise, hold cash (or a money market fund). We evaluate lookback windows of 1 to 22 days and report results for the 10-day specification, which balances signal quality against transaction costs: the Taiwan Sharpe ratio improves by 29% (from 1.141 to 1.473) with 38% fewer switches per year (29 versus 47). We acknowledge that the lookback selection introduces a degree of data mining; however, all eight lookback windows from 1 to 22 days produce positive net Sharpe ratios, and four of eight exceed the Harvey threshold.

Return measurement caveat. A critical distinction arises between close-to-close (c2c) and open-to-open (o2o) return measurement. The standard c2c return includes the overnight opening gap, which in Asian markets mechanically reflects the previous day's U.S. session. This gap alone accounts for approximately 78% of the c2c strategy alpha ($R^2_{\text{gap}} = 0.35$ versus $R^2_{\text{c2c}} = 0.18$, with intraday reversal). Since the opening gap is realized before a trader can act on the signal, the c2c Sharpe ratio overstates implementable performance. We therefore report both c2c and o2o results: c2c results constitute a *non-implementable* benchmark that measures the total information transmission across markets, while o2o results measure the *implementable* portion.

A.2 Main Results

Table 12 presents the time-zone momentum results for Taiwan, Japan, and combined portfolios, reporting both close-to-close (c2c) and open-to-open (o2o) Sharpe ratios.

Table 12: Time-Zone Momentum Strategy Performance: Close-to-Close vs. Open-to-Open

Strategy	Sharpe (c2c)	Sharpe (o2o)	t-stat (o2o)	MDD (%)	Sw/yr	Period	Harvey?
Panel A: Individual Markets (10-day SPY Momentum)							
Taiwan (0050.TW)	1.473	0.87	2.22	-12.8	29	2012–2025	No
Japan (Nikkei 225)	1.306	0.78	2.00	-14.5	28	2012–2025	No
Panel B: Combinations (c2c-based, see caveat)							
TW + JP 50/50	1.810	—	—	-11.9	—	2012–2025	—
Global (US VT + TW TZ)	1.610	—	—	-8.4	—	2012–2025	—
Panel C: Controls							
EWT (US-listed TW ETF)	—	—	< 1.96	—	—	2012–2025	No
India (Sensex)	—	—	< 1.96	—	—	2012–2025	No
Indonesia (JCI)	—	—	< 1.96	—	—	2012–2025	No

Notes: "c2c" denotes close-to-close returns (including overnight gaps); "o2o" denotes open-to-open returns (implementable portion only). The c2c Sharpe ratios are inflated because the overnight opening gap—which mechanically reflects the previous U.S. session—is captured in c2c returns but cannot be traded upon, as it is realized before the signal can be acted on. All t-statistics in the "t-stat (o2o)" column use Newey-West HAC standard errors with automatic bandwidth selection. The Harvey et al. [2016] threshold is $t > 3.0$. Transaction costs of 0.186% per switch are applied throughout.

Using close-to-close returns, the Taiwan 10-day SPY momentum strategy generates a net Sharpe ratio of 1.473 ($t = 3.76$), exceeding the Harvey et al. [2016] threshold. When measured using open-to-open returns, the Sharpe ratio drops to 0.87 ($t = 2.22$), below the Harvey threshold. Approximately 78% of the c2c alpha is absorbed by the opening gap.

The c2c results confirm a strong and persistent information transmission channel from U.S. to Asian markets ($r = 0.376$), with six Asia-Pacific markets exceeding the Harvey threshold on a c2c basis (Hong Kong $t = 4.12$, Australia $t = 4.04$, Singapore $t = 4.03$, Korea $t = 3.83$, Taiwan $t = 3.76$, Japan $t = 3.69$). Rather than interpreting the c2c–o2o gap as a failure, we view it as evidence of efficient price discovery: the Asian opening auction incorporates approximately 78% of overnight U.S. information within the first transaction of the trading day.

A.3 Overnight Gap as the Primary Channel

A granular decomposition of 0050.TW returns reveals that the overnight opening gap accounts for 87% of total close-to-close returns over the 2012–2025 sample. When conditioned on the previous day's SPY return sign, the gap is strongly asymmetric: following SPY up-days, the average 0050.TW opening gap is +10.73 basis points per day ($t = 6.845$), whereas following SPY down-days, the gap averages −8.91 bp ($t = -5.23$). The intraday (open-to-close) component shows no significant SPY-conditional pattern ($t = 0.82$), confirming that cross-market information is incorporated almost entirely through the opening auction.

Figure 3 visualizes the two key mechanisms. Panel (a) shows the distribution of 0050.TW overnight gaps conditioned on the previous day's SPY return direction. Panel (b) documents the positive relationship between VIX levels and next-day 0050.TW absolute returns, confirming the VIX-to-Taiwan volatility transmission channel that underpins the 8.63/VIX strategy.

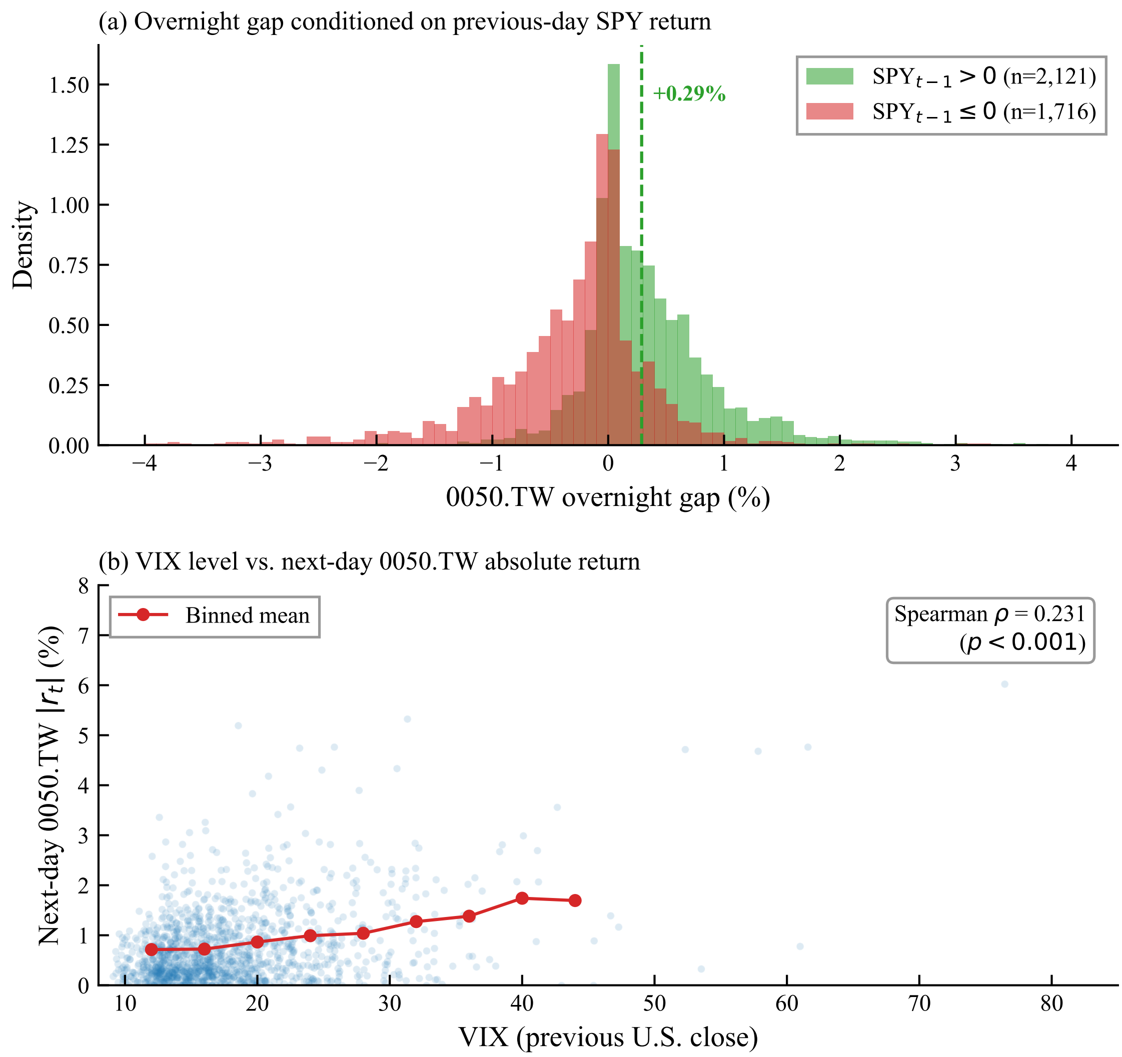

Figure 3: Information transmission from U.S. to Taiwan equity markets, 2010–2026. Panel (a): Distribution of 0050.TW overnight gaps conditioned on the sign of the previous day's SPY close-to-close return. Panel (b): VIX level versus next-day 0050.TW absolute return. Red circles show binned means.

A.4 Identification: Time-Zone Gap, Not Factor Exposure

We confirm that the cross-market predictability reflects genuine information transmission rather than common factor exposure using three tests: (1) the iShares MSCI Taiwan ETF (EWT), which trades on the NYSE during U.S. hours, fails to achieve statistical significance ($t < 1.96$), confirming the alpha originates from the time-zone gap; (2) European close-to-U.S.-open signals show a negative correlation ($r = -0.07$), consistent with the zero-overlap requirement; and (3) India and Indonesia controls both fail the $t > 3.0$ threshold.

A.5 Robustness

The c2c information transmission channel is robust to: multi-period out-of-sample tests (positive Sharpe in all five sub-periods, combined $t = 4.47$–4.60), block bootstrap (95% CI [0.65, 2.24], excluding zero), Bonferroni correction (six markets, adjusted threshold $t > 3.3$; all six pass on c2c basis), transaction cost sensitivity (breakeven > 3% per switch for c2c), and lookback sensitivity (5-day and 10-day both exceed Harvey threshold on c2c). On an o2o basis, none of the individual markets pass the Bonferroni-adjusted threshold, underscoring the gap between information content and implementable alpha.

A.6 ETFs Versus Individual Stocks

The time-zone momentum strategy performs more consistently on the 0050.TW ETF than on individual Taiwanese stocks. Of the securities tested, four pass the Harvey et al. [2016] $t > 3.0$ threshold on a c2c basis: 0056.TW ($t = 3.44$), Hon Hai ($t = 3.40$), 0050.TW ($t = 3.25$), and Cathay Financial ($t = 3.06$). ETFs offer a more stable substrate because diversification smooths idiosyncratic noise and the ETF mechanically reflects the broad market's response to U.S. information.

References

Acerbi, C. & Szekely, B. (2014). Back-testing expected shortfall. *Risk*, 27(11), 76–81.

Ang, A. & Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443–494.

Barber, B.M., Lee, Y.-T., Liu, Y.-J., & Odean, T. (2009). Just how much do individual investors lose by trading? *Review of Financial Studies*, 22(2), 609–632.

Barclay, M.J. & Hendershott, T. (2003). Price discovery and trading after hours. *Review of Financial Studies*, 16(4), 1041–1073.

Barndorff-Nielsen, O.E., Kinnebrock, S., & Shephard, N. (2010). Measuring downside risk—realized semivariance. In T. Bollerslev, J. Russell, & M. Watson (Eds.), *Volatility and Time Series Econometrics: Essays in Honor of Robert Engle* (pp. 117–136). Oxford University Press.

Black, F. (1976). Studies of stock price volatility changes. *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economic Statistics Section*, 177–181.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Bollerslev, T. & Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.

Bozovic, M. (2024). VIX-managed portfolios. *International Review of Financial Analysis*, 95, 103353.

Christie, A.A. (1982). The stochastic behavior of common stock variances: Value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.

Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.

Diebold, F.X. & Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.

Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

Engle, R.F., Ghysels, E., & Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *Review of Economics and Statistics*, 95(3), 776–797.

Eun, C.S. & Shim, S. (1989). International transmission of stock market movements. *Journal of Financial and Quantitative Analysis*, 24(2), 241–256.

Fissler, T. & Ziegel, J.F. (2016). Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.

Fleming, J., Kirby, C., & Ostdiek, B. (2001). The economic value of volatility timing. *Journal of Finance*, 56(1), 329–352.

Gagnon, L. & Karolyi, G.A. (2010). Multi-market trading and arbitrage. *Journal of Financial Economics*, 97(1), 53–80.

Glosten, L.R., Jagannathan, R., & Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.

Hamao, Y., Masulis, R.W., & Ng, V. (1990). Correlations in price changes and volatility across international stock markets. *Review of Financial Studies*, 3(2), 281–307.

Harvey, C.R., Liu, Y., & Zhu, H. (2016). …and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

Harvey, C.R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M., & Van Hemert, O. (2018). The impact of volatility targeting. *Journal of Portfolio Management*, 45(1), 14–33.

Hwang, S. & Valls Pereira, P.L. (2006). Small sample properties of GARCH estimates and persistence. *European Journal of Finance*, 12(6–7), 473–494.

JPMorgan. (1996). *RiskMetrics—Technical Document* (4th ed.). New York: J.P. Morgan/Reuters.

Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.

Lin, W.-L., Engle, R.F., & Ito, T. (1994). Do bulls and bears move across borders? International transmission of stock returns and volatility. *Review of Financial Studies*, 7(3), 507–538.

Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.

Nelson, D.B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370.

Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

Rapach, D.E., Strauss, J.K., & Zhou, G. (2013). International stock return predictability: What is the role of the United States? *Journal of Finance*, 68(4), 1633–1662.

Whaley, R.E. (2000). The investor fear gauge. *Journal of Portfolio Management*, 26(3), 12–17.

Whaley, R.E. (2009). Understanding the VIX. *Journal of Portfolio Management*, 35(3), 98–105.